

Interim Report: Fixed-Term Rental Agreements in Southern Austria

Group 16

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1 Setup

```
load_packages <- function(packages) {
  lapply(packages, function(pkg) {
    if (!require(pkg, character.only = TRUE)) {
      install.packages(pkg)
      library(pkg, character.only = TRUE)
    }
  })
}
```

```

}

load_packages(c(
  "pander",
  "car",
  "vcd",
  "effects",
  "carData",
  "dplyr",
  "DHARMa",
  "performance",
  "see",
  "ggeffects",
  "ggplot2",
  "patchwork"
))

panderOptions('table.alignment.default', 'right')

```

2 Introduction

The data used in this report stem from the Austrian Microcensus, a quarterly sample survey carried out by Statistik Austria. The full survey is based on randomly selected households and usually includes information on around 40000 to 50000 persons. However for this project only a subset of the dataset was considered.

This report analyzes the probability of having a fixed-term rental contract in Southern Austria (NUTS-1 region 2: Carinthia and Styria) on the basis of Austrian Microcensus data from 2012. The dependent variable is `wfrist`, indicating whether the respondent has a fixed-term rental agreement or not (yes/no). As explanatory variables, two metric predictors are considered: `dstd` (weekly working hours) and `kjahr` (years since completion of the highest level of education). In addition, two categorical predictors are included: `bfst` - (family status: single, married, widowed/divorced) and `xeinw` - (community size: up to 2000 (small-S), 2001-10000 (medium-M), 10001-100000 (large-L), 100001+ (very large-VL)).

The predictors were selected because they capture different characteristics that may be related to rental contract type. Weekly working hours (`dstd`) reflect the individuals's labor-market situation, years since education completion (`kjahr`) indicate their position in the life course, family status (`bfst`) represents the household and partnership situation and community size (`xeinw`) captures differences in the local housing context.

Because the dependent variable is binary, the subsequent inferential analysis will rely on **binary logistic regression**.

2.1 Research questions and hypotheses

The primary research question of this analysis is: **Which factors predict whether a renter in Southern Austria has a fixed-term (limited-duration) or permanent (open-ended) rental agreement?**

We hypothesize that rental contract type is associated with:

- **Working hours** (`dstd`, weekly working hours): People with higher weekly working hours may be more likely to have permanent rental contracts (`wfrist = no`), as regular employment can be associated with greater economic stability and more long-term housing arrangements. By contrast, respondents with lower weekly working hours may be more likely to live in fixed-term rental arrangements (`wfrist = yes`).
- **Year since educational completion** (`kjahr`, years since completing highest education): Individuals who completed their education longer ago may be more established in their life course and therefore more likely to have permanent rental contracts. In contrast, respondents who completed their education more recently may still be in a more flexible or transitional phase and thus more likely to have fixed-term agreements.
- **Family situation** (`bfst`, family status): Married individuals may be more likely to have permanent rental agreements, as long-term household stability may increase the preference for durable housing arrangements. Single and widowed/divorced respondents, by contrast, may more often have fixed-term contracts due to greater mobility or less stable living situations.
- **Community size** (`xeinw`, size of community of residence): People living in larger communities may be more likely to have fixed-term rental agreements, since housing markets in more populous areas may be more dynamic and less stable than in smaller communities.

2.2 Starting point and objectives

This report focuses on **data management and descriptive statistics** as the first phase of the regression project. The objectives are:

- Characterize the sample of renters in Southern Austria with respect to all five analysis variables
- Document relationships between predictors and the response variable
- Identify potential interactions among predictors
- Prepare for subsequent binary logistic regression modeling in the final report

```

variable_overview <- data.frame(
  Group = c(
    "Response",
    "Predictors (metric)",
    "Predictors (metric)",
    "Predictors (categorical)",
    "Predictors (categorical)"
  ),
  Variable = c(
    "wfrist",
    "dstd",
    "kjahr",
    "bfst",
    "xeinw"
  ),
  Scale = c(
    "Nominal",
    "Ratio",
    "Ratio",
    "Nominal",
    "Ordinal"
  ),
  Type = c(
    "Binary",
    "Metric",
    "Metric",
    "Polytomous",
    "Polytomous"
  ),
  "Categories/Range" = c(
    "yes, no",
    "weekly working hours",
    "years since completing highest education",
    "3 levels (single, married, widowed/divorced)",
    "4 levels (up to 2000, 2001-10000, 10001-100000, 100001+)"
  ),
  "Missing Values" = c(
    "Excluded",
    "Excluded",
    "Excluded",
    "Excluded",
    "Excluded"
  ),
)

```

```

    check.names = FALSE
  )

  pander(
    variable_overview,
    caption = paste(
      "Overview of the response and predictor variables used in the study,",
      "including measurement scale, variable type, category structure or range,",
      "and treatment of missing values."
    )
  )
)

```

Table 1: Overview of the response and predictor variables used in the study, including measurement scale, variable type, category structure or range, and treatment of missing values. (continued below)

Group	Variable	Scale	Type
Response	wfrist	Nominal	Binary
Predictors (metric)	dstd	Ratio	Metric
Predictors (metric)	kjahr	Ratio	Metric
Predictors (categorical)	bfst	Nominal	Polytomous
Predictors (categorical)	xeinw	Ordinal	Polytomous

Categories/Range	Missing Values
yes, no	Excluded
weekly working hours	Excluded
years since completing highest education	Excluded
3 levels (single, married, widowed/divorced)	Excluded
4 levels (up to 2000, 2001-10000, 10001-100000, 100001+)	Excluded

2.3 Data Preparation

The dataset was imported and prepared for the analysis by recoding all -3 values as missing values NA. The dataset was then restricted to respondents from Southern Austria (`xnuts` = 2) and only observations with valid information on the dependent variable `wfrist` were retained. The variables used in the analysis were recoded into interpretable factor categories. In addition, only the variables relevant for the analysis were kept and remaining missing values were removed.

```

mc <- read.csv(
  "D:/Documents/00-Personal/01-FHTW/SS2026/MDS-BB-2-SS2026-MA-EN/20260417_Regression Project,
  na.strings = c("NA", "")
)

# Replace -3 values with NA
mc <- mc %>%
  mutate(across(everything(), ~ replace(.x, .x == -3, NA)))

# Subset to Southern Austria
mc_south <- subset(mc, xnuts1 == 2)

# Keep only rows where wfrist is valid
mc_south <- subset(mc_south, !is.na(wfrist))

# Recode wfrist as factor with meaningful labels
mc_south$wfrist <- factor(
  mc_south$wfrist,
  levels = c(1, 2),
  labels = c("yes", "no")
)

# Recode categorical predictors as factors with meaningful labels
mc_south$bfst <- factor(
  mc_south$bfst,
  levels = 1:4,
  labels = c("single", "married", "widowed", "divorced")
)

mc_south$xeinw <- factor(
  mc_south$xeinw,
  levels = 1:4,
  labels = c("up to 2000", "2001-10000", "10001-100000", "100001+")
)

# Select analysis variables
mc_analysis <- mc_south[, c("wfrist", "dstd", "kjahr", "bfst", "xeinw")]

# Remove remaining NAs
mc_analysis <- na.omit(mc_analysis)

# Final sample size
n_total <- nrow(mc_analysis)

```

```

n_fixed_term <- sum(mc_analysis$wfrist == "yes")
n_permanent <- sum(mc_analysis$wfrist == "no")
pct_fixed_term <- round(100 * n_fixed_term / n_total, 1)

# Community size distribution (xeinw)
n_up_to_2000 <- sum(mc_analysis$xeinw == "up to 2000")
n_2001_10000 <- sum(mc_analysis$xeinw == "2001-10000")
n_10001_100000 <- sum(mc_analysis$xeinw == "10001-100000")
n_100001_plus <- sum(mc_analysis$xeinw == "100001+")

# Family status distribution (bfst)
n_single <- sum(mc_analysis$bfst == "single")
n_married <- sum(mc_analysis$bfst == "married")
n_widowed <- sum(mc_analysis$bfst == "widowed")
n_divorced <- sum(mc_analysis$bfst == "divorced")

```

```

pander(
  head(mc_analysis),
  caption = "First six observations of the dataset."
)

```

Table 2: First six observations of the dataset.

	wfrist	dstd	kjahr	bfst	xeinw
1	yes	30	26	divorced	100001+
88	no	25	25	divorced	2001-10000
90	no	40	16	single	10001-100000
114	no	30	12	single	2001-10000
160	no	30	30	married	100001+
161	no	70	15	married	100001+

```

pander(
  cbind(
    Frequency = table(mc_analysis$xeinw),
    Percent = round(prop.table(table(mc_analysis$xeinw)) * 100, 1)
  ),
  caption = "Distribution of community size (xeinw)."
)

```


Table 3: Distribution of community size (xeinw).

	Frequency	Percent
up to 2000	21	8.3
2001-10000	83	32.8
10001-100000	118	46.6
100001+	31	12.3

```
pander(
  cbind(
    Frequency = table(mc_analysis$bfst),
    Percent = round(prop.table(table(mc_analysis$bfst)) * 100, 1)
  ),
  caption = "Distribution of family status (bfst) before recoding."
)
```

Table 4: Distribution of family status (bfst) before recoding.

	Frequency	Percent
single	136	53.8
married	72	28.5
widowed	1	0.4
divorced	44	17.4

Since the category **widowed** contained only one observation, it was combined with **divorced** in order to avoid sparse categories and to ensure a meaningful analysis.

```
mc_analysis$bfst <- as.character(mc_analysis$bfst)

mc_analysis$bfst[mc_analysis$bfst == "widowed"] <- "widowed/divorced"
mc_analysis$bfst[mc_analysis$bfst == "divorced"] <- "widowed/divorced"

mc_analysis$bfst <- factor(
  mc_analysis$bfst,
  levels = c("single", "married", "widowed/divorced")
)
```

```
pander(
  cbind(
```

```

Frequency = table(mc_analysis$bfst),
Percent = round(prop.table(table(mc_analysis$bfst)) * 100, 1)
),
caption = "Distribution of family status (bfst) after recoding."
)

```

Table 5: Distribution of family status (bfst) after recoding.

	Frequency	Percent
single	136	53.8
married	72	28.5
widowed/divorced	45	17.8

After data preparation the analysis dataset contains 253 observations. Of these:

- **Contract type:** 41 have fixed-term rental agreements; 212 have permanent agreements.
- **Community size:** 21 live in small communities with up to 2000 inhabitants, 83 in medium communities with 2001 – 10000 inhabitants, 118 in large communities with 10001 – 100000 inhabitants and 31 in very large communities with more than 100000 inhabitants.
- **Family status:** 136 single individuals, 72 married and 45 are widowed/divorced.

The analysis includes all four community sizes, enabling comparison of rental contract preferences across different groups in Southern Austria. Family status is dominated by married and single respondents, with a small combined category of widowed/divorced. This distribution allows for meaningful analysis of how family status relates to rental contract type while avoiding issues with sparse categories.

Sample size: The subset relevant for our analysis consists of 253 individuals.

3 Descriptive Analysis

3.1 Univariate Analysis

3.1.1 Response Variable: **wfrist (Fixed-Term Rental Agreement)**

```
# Frequency table
wfrist_freq <- table(mc_analysis$wfrist)
pander(
  wfrist_freq,
  caption = "Frequency distribution: Fixed-term rental agreement"
)
```

Table 6: Frequency distribution: Fixed-term rental agreement

yes	no
41	212

Distribution of Fixed-Term vs. Permanent Rental Agreements. (N = 253) Bar chart showing 41 respondents (16.2%) with fixed-term agreements and 212 (83.8%) with permanent agreements.

```
# Proportion
wfrist_prop <- prop.table(table(mc_analysis$wfrist))
pander(wfrist_prop, caption = "Proportion: Fixed-term rental agreement")
```

Table 7: Proportion: Fixed-term rental agreement

yes	no
0.1621	0.8379

Distribution of Fixed-Term vs. Permanent Rental Agreements. (N = 253) Bar chart showing 41 respondents (16.2%) with fixed-term agreements and 212 (83.8%) with permanent agreements.

```
# Barplot
barplot(
  wfrist_freq,
  main = sprintf(
    "Distribution of Fixed-Term Rental Agreements (N = %d)",
    n_total
  ),
  xlab = "Fixed-term contract",
  ylab = "Frequency",
  col = c("steelblue", "coral")
)
```

Distribution of Fixed-Term Rental Agreements (N = 253)

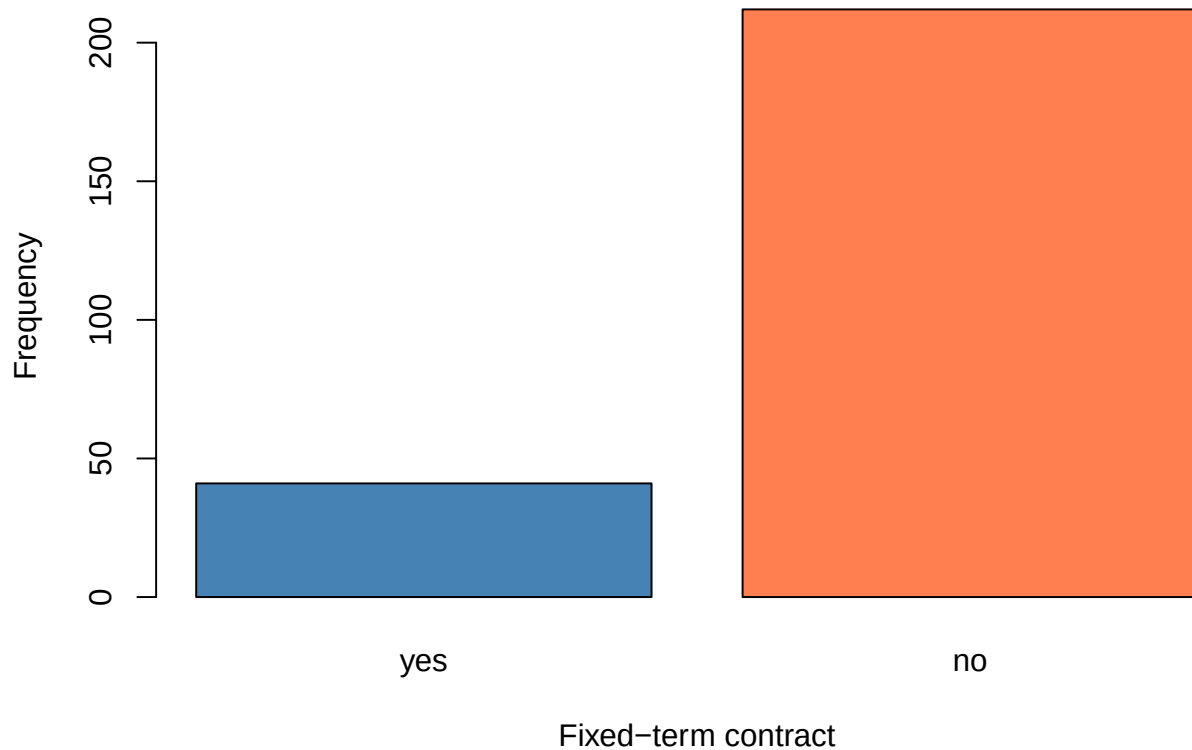


Figure 1: Distribution of Fixed-Term vs. Permanent Rental Agreements. (N = 253) Bar chart showing 41 respondents (16.2%) with fixed-term agreements and 212 (83.8%) with permanent agreements.

The distribution chart shows that the response variable `wfrist` is unequally distributed. Only 41 respondents (16.2%) have a fixed-term rental contract, while 212 respondents (83.8%) have a permanent contract. Thus, permanent rental agreements clearly dominate in the sample.

3.1.2 Metric Predictor 1: `dstd` (Weekly Working Hours)

```
# Summary statistics
summary_dstd <- summary(mc_analysis$dstd)
dstd_sd <- sd(mc_analysis$dstd)
cap_dstd <- sprintf(
  "Summary statistics for weekly working hours (dstd). Range %.0f-%.0f hours/week; mean %.2f",
  summary_dstd[["Min."]],
  summary_dstd[["Max."]],
```

```
summary_dstd[["Mean"]],
summary_dstd[["Median"]],
dstd_sd
)
pander(data.frame(t(summary_dstd)), caption = cap_dstd)
```

Table 8: Summary statistics for weekly working hours (dstd). Range 4-80 hours/week; mean 36.84; median 40; SD 11.67 hours.

Var1	Var2	Freq
A	Min.	4
A	1st Qu.	30
A	Median	40
A	Mean	36.84
A	3rd Qu.	40
A	Max.	80

```
# Visualization
par(mfrow = c(1, 2))
hist(
  mc_analysis$dstd,
  probability = TRUE,
  main = "Distribution of Weekly \nWorking Hours",
  xlab = "Hours per week",
  ylab = "Frequency",
  col = "steelblue",
  breaks = 20
)
lines(density(mc_analysis$dstd), lwd = 1)
lines(density(mc_analysis$dstd, bw = 5), lwd = 2)
boxplot(
  mc_analysis$dstd,
  main = "Boxplot: Weekly Working Hours",
  xlab = "Hours per week",
  horizontal = TRUE,
  col = "steelblue"
)
```

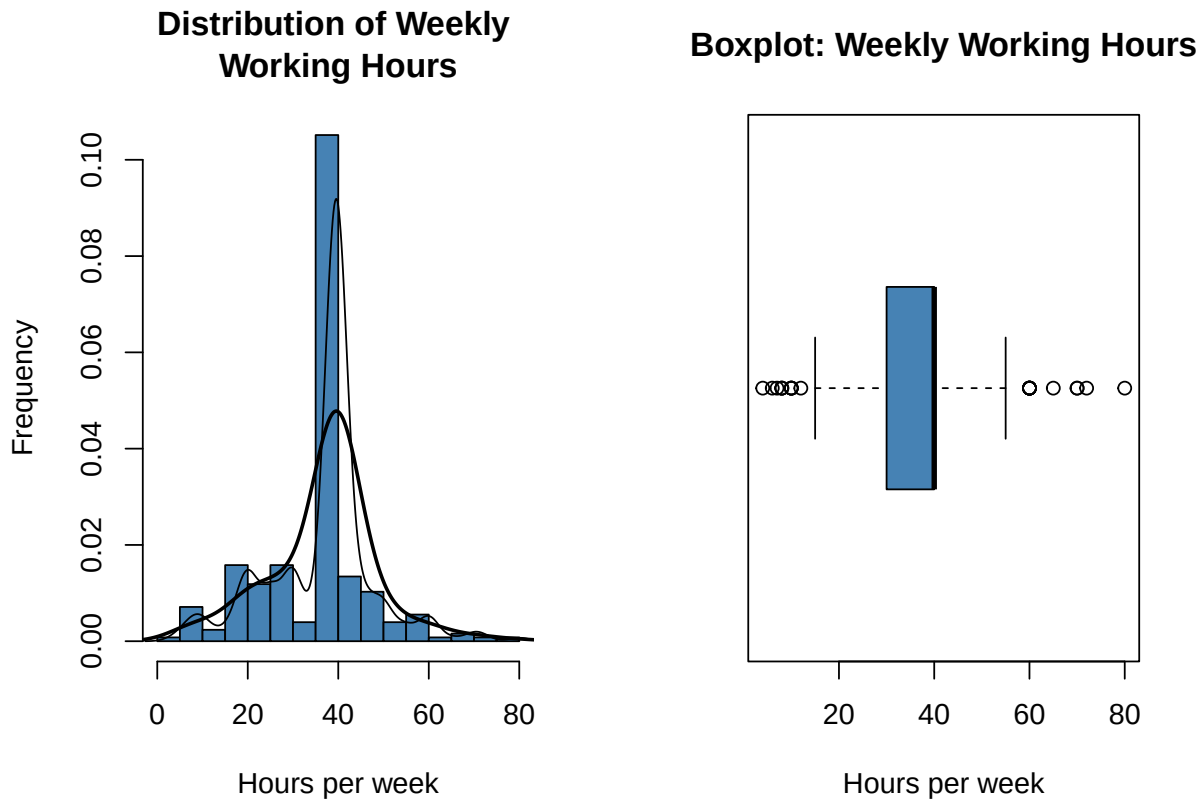


Figure 2: Distribution of Weekly Working Hours. (a) Histogram with density curves centered around 40 hours/week; (b) Boxplot showing outliers at both low (4-14 hours) and high (60-80 hours) employment intensity.

```
par(mfrow = c(1, 1))
```

Weekly working hours range from 4 to 80 hours per week. The mean is 36.84 hours, while the median is 40 hours, indicating that the distribution is centered around full-time employment, as also shown in the histogram with the smoothed density curve. Since the mean is slightly lower than the median, the distribution appears to be somewhat left-skewed, which may be caused by respondents with relatively low weekly working hours. In addition, the standard deviation of 11.67 hours suggests a moderate spread around the center.

Based on the boxplot, the distribution of weekly working hours contains several potential outliers. These occur both among very low and very high values, indicating that a small number of respondents report working hours that differ considerably from the majority of the sample.

```
bp <- boxplot(mc_analysis$dstd, plot = FALSE)
lower_outlier_rows <- mc_analysis[mc_analysis$dstd < bp$stats[1], ]
```

```
upper_outlier_rows <- mc_analysis[mc_analysis$dstd > bp$stats[5], ]

cap_lower <- sprintf(
  "Lower outliers for weekly working hours. Respondents with working hours below %.1f hours/v",
  bp$stats[1],
  nrow(lower_outlier_rows)
)

pander(lower_outlier_rows, caption = cap_lower)
```

Table 9: Lower outliers for weekly working hours. Respondents with working hours below 15.0 hours/week (N = 11). All observations with complete information on all variables.

	wfrist	dstd	kjahr	bfst	xeinw
887	no	4	28	married	100001+
921	no	12	34	married	up to 2000
2002	yes	6	6	single	100001+
3221	no	8	5	single	10001-100000
3396	no	7	42	married	2001-10000
3553	yes	10	30	married	10001-100000
4297	no	10	2	single	2001-10000
4575	no	10	7	single	2001-10000
8235	no	8	43	married	up to 2000
8512	no	8	9	married	10001-100000
9024	no	10	29	single	10001-100000

The lower outliers reflect respondents with unusually low weekly working hours. They are distributed across different categories of family status and community size and are mostly associated with permanent rental contracts. This suggests that these low values do not form a distinct subgroup but rather represent individual cases of comparatively low employment intensity.

```
cap_upper <- sprintf(
  "Upper outliers for weekly working hours. Respondents with working hours above %.1f hours/v",
  bp$stats[5],
  nrow(upper_outlier_rows)
)

pander(upper_outlier_rows, caption = cap_upper)
```

Table 10: Upper outliers for weekly working hours. Respondents with working hours above 55.0 hours/week (N = 12). All observations with complete information on all variables.

	wfrist	dstd	kjahr	bfst	xeinw
161	no	70	15	married	100001+
791	yes	72	7	single	100001+
1922	no	70	20	single	100001+
2853	no	60	29	married	10001-100000
3665	yes	60	44	widowed/divorced	10001-100000
3846	no	60	24	single	10001-100000
4296	no	65	16	single	2001-10000
4362	no	60	2	single	10001-100000
4856	no	60	24	widowed/divorced	10001-100000
5491	no	60	17	single	2001-10000
8032	no	80	17	single	10001-100000
8223	no	60	11	single	2001-10000

The upper outliers reflect unusually high weekly working hours and are distributed across several categories of family status and community size. Most of these cases are linked to permanent rental contracts, indicating that very high working hours do not primarily characterize respondents with fixed-term agreements.

Overall, these outliers suggest variation in working intensity rather than a clearly distinct subgroup.

3.1.3 Metric Predictor 2: kjahr (Years Since Completing Education)

```
# Summary statistics
summary_kjahr <- summary(mc_analysis$kjahr)
kjahr_sd <- sd(mc_analysis$kjahr)
cap_kjahr <- sprintf(
  "Summary statistics for years since education completion (kjahr). Range %.0f-%.0f years; m
  summary_kjahr[["Min."]],
  summary_kjahr[["Max."]],
  summary_kjahr[["Mean"]],
  summary_kjahr[["Median"]],
  kjahr_sd
)
pander(data.frame(t(summary_kjahr)), caption = cap_kjahr)
```


Table 11: Summary statistics for years since education completion (kjahr). Range 0-44 years; mean 18.84; median 19; SD 11.69 years.

Var1	Var2	Freq
A	Min.	0
A	1st Qu.	8
A	Median	19
A	Mean	18.84
A	3rd Qu.	28
A	Max.	44

```
# Visualization
par(mfrow = c(1, 2))
hist(
  mc_analysis$kjahr,
  probability = TRUE,
  xlab = "Years",
  ylab = "Frequency",
  col = "lightgreen",
  main = "Distribution of Years Since \nEducation Completion",
  breaks = 25
)
lines(density(mc_analysis$kjahr), lwd = 1)
lines(density(mc_analysis$kjahr, bw = 5), lwd = 2)

boxplot(
  mc_analysis$kjahr,
  xlab = "Years",
  horizontal = TRUE,
  main = "Boxplot: Years Since \nEducation Completion",
  col = "lightgreen"
)
```

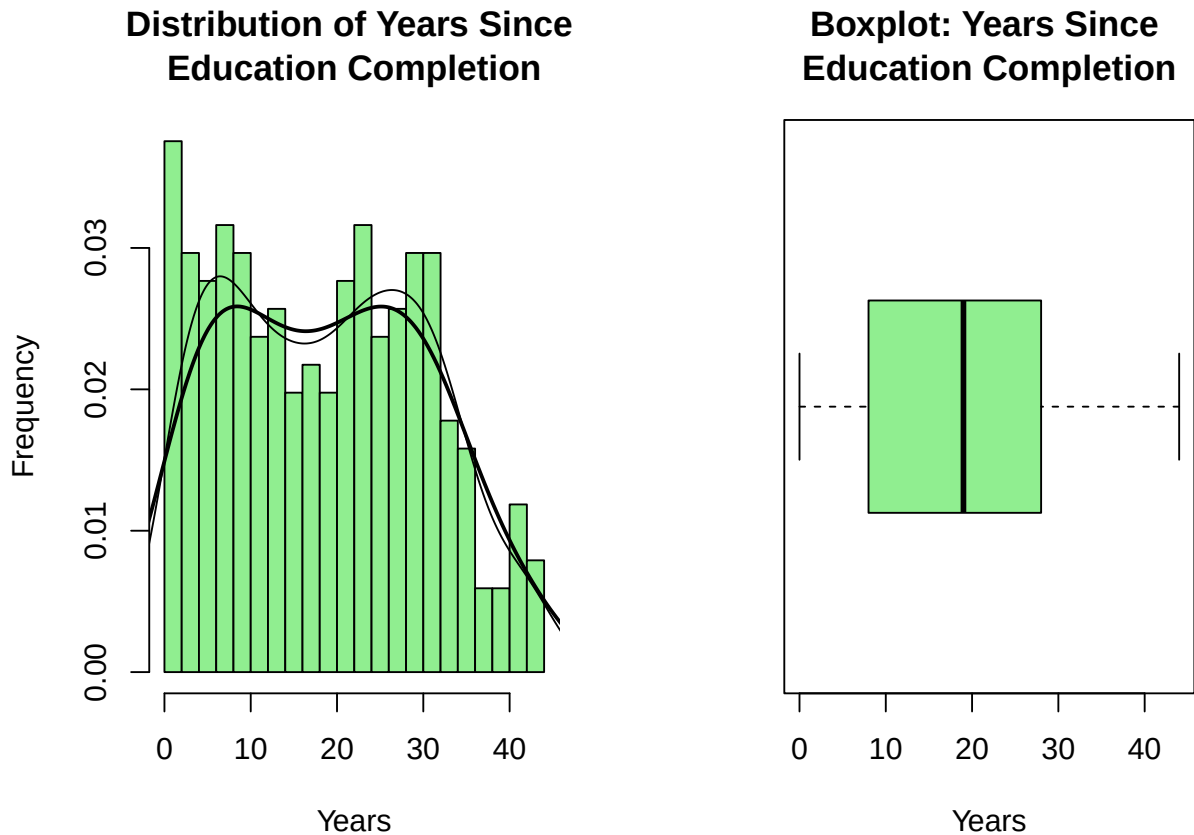


Figure 3: Distribution of Years Since Completing Education. (a) Histogram with density curves showing a broad distribution across approximately 0-44 years, with mild clustering in the lower and middle ranges and fewer observations at higher values; (b) Boxplot indicating moderate variability, a median around 19 years and no pronounced outliers.

```
par(mfrow = c(1, 1))
```

The distribution of years since completing education is broad, covering approximately 0 to 44 years and is not concentrated within a narrow interval. The histogram suggests a fairly wide spread of values, with mild clustering in the lower and middle ranges and a gradual decline toward the higher end. The density curves indicate slight irregularities rather than a perfectly symmetric shape. The boxplot supports this interpretation, showing moderate variability, a median around 19 years and no pronounced outliers. Overall, the variable reflects substantial heterogeneity in time since educational completion, without evidence of extreme or highly unusual values.

3.1.4 Categorical Predictor 1: bfst (Family Status)

```
# Frequency table
bfst_freq <- table(mc_analysis$bfst)
cap_bfst_freq <- sprintf(
  "Frequency distribution of family status. Single: %d; married: %d; widowed/divorced: %d (N
  bfst_freq[1],
  bfst_freq[2],
  bfst_freq[3],
  sum(bfst_freq)
)
pander(bfst_freq, caption = cap_bfst_freq)
```

Table 12: Frequency distribution of family status. Single: 136; married: 72; widowed/divorced: 45 (N = 253).

single	married	widowed/divorced
136	72	45

```
# Proportion
bfst_prop <- prop.table(bfst_freq)
cap_bfst_prop <- sprintf(
  "Proportion of family status categories. Single: %.1f%%; married: %.1f%%; widowed/divorced
  bfst_prop[[1]] * 100,
  bfst_prop[[2]] * 100,
  bfst_prop[[3]] * 100
)
pander(bfst_prop, caption = cap_bfst_prop)
```

Table 13: Proportion of family status categories. Single: 53.8%; married: 28.5%; widowed/divorced: 17.8%.

single	married	widowed/divorced
0.5375	0.2846	0.1779

```
# Barplot
barplot(
  bfst_freq,
  xlab = "Family status",
```

```
ylab = "Frequency",
col = c("lightsteelblue", "lightcoral", "lightyellow")
)
```

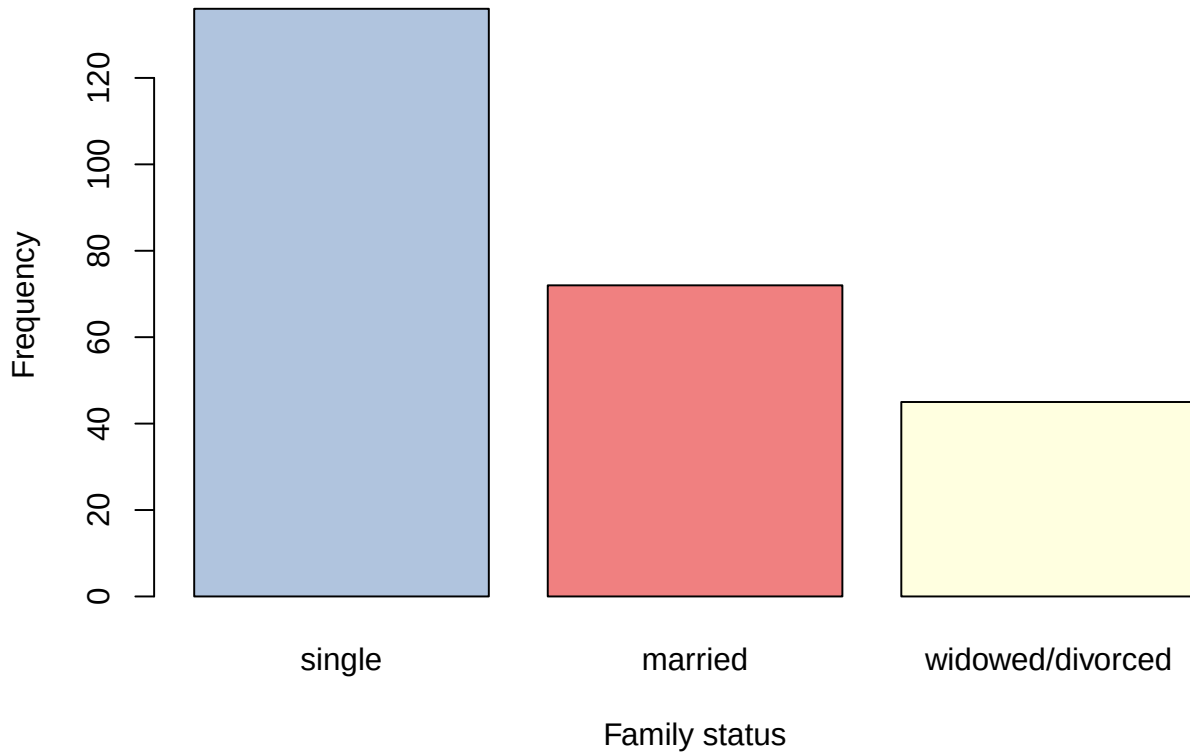


Figure 4: Distribution of Family Status Among Southern Austrian Renters. Single: 136 (53.8%), married: 72 (28.5%), widowed/divorced: 45 (17.8%).

Family status is unevenly distributed in the sample. The majority of respondents are single (53.8%), followed by married respondents (28.5%), while widowed/divorced respondents (17.8%) make up the smallest share.

3.1.5 Categorical Predictor 2: `xeinw` (Community size Classification)

```
# Frequency table
xeinw_freq <- table(mc_analysis$xeinw)
cap_xeinw_freq <- sprintf(
  "Frequency distribution of community size. Small (up to 2000): %d; medium (2001-10000): %d\n",
  xeinw_freq[1],
  xeinw_freq[2])
```

```

xeinw_freq[2],
xeinw_freq[3],
xeinw_freq[4],
sum(xeinw_freq)
)
pander(xeinw_freq, caption = cap_xeinw_freq)

```

Table 14: Frequency distribution of community size. Small (up to 2000): 21; medium (2001-10000): 83; large (10001-100000): 118; very large (>100000): 31 (N = 253).

up to 2000	2001-10000	10001-100000	100001+
21	83	118	31

```

# Proportion
xeinw_prop <- prop.table(xeinw_freq)
cap_xeinw_prop <- sprintf(
  "Proportion of community size categories. Small %.1f%%, medium %.1f%%, large %.1f%%, very l
  xeinw_prop[[1]] * 100,
  xeinw_prop[[2]] * 100,
  xeinw_prop[[3]] * 100,
  xeinw_prop[[4]] * 100
)
pander(xeinw_prop, caption = cap_xeinw_prop)

```

Table 15: Proportion of community size categories. Small 8.3%, medium 32.8%, large 46.6%, very large 12.3%.

up to 2000	2001-10000	10001-100000	100001+
0.083	0.3281	0.4664	0.1225

```

# Barplot
bp <- barplot(
  xeinw_freq,
  ylab = "Frequency",
  col = c("steelblue", "coral", "gold", "darkseagreen"),
  names.arg = FALSE
)

text(

```

```

x = bp,
y = par("usr")[3] - 5,
labels = names(xeinw_freq),
srt = 45,
adj = 1,
xpd = TRUE
)

```

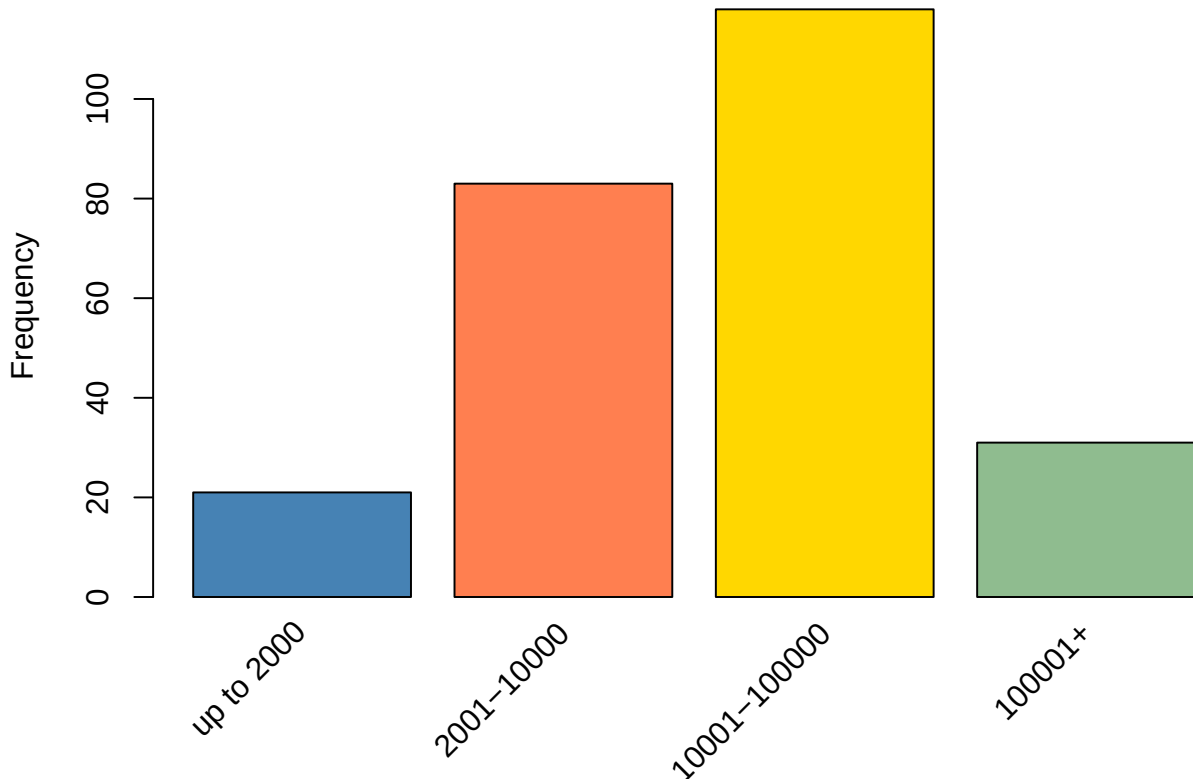


Figure 5: Distribution of Community Size Classification Among Southern Austrian Renters. Bar chart showing large communities (10001-100000) house 46.6% of renters, medium (2001-10000) 32.8%, very large (>100000) 12.3%, small (up to 2000) 8.3%.

Community size is unevenly distributed in the sample. Nearly half of the respondents live in communities with 10001-100000 inhabitants (46.6%), while about one third live in communities with 2001-10000 inhabitants (32.8%). Only smaller proportions come from communities with more than 100000 inhabitants (12.3%) or up to 2000 inhabitants (8.3%).

3.2 Bivariate Analysis: Predictor-Response Relationships

3.2.1 dstd (Weekly Hours) vs. wfrist (Fixed-Term Agreement)

```
# Spinogram for metric predictor against binary response
spineplot(
  mc_analysis$wfrist ~ mc_analysis$dstd,
  xlab = "Weekly working hours (dstd)",
  ylab = "Fixed-term contract",
  breaks = 5
)
```

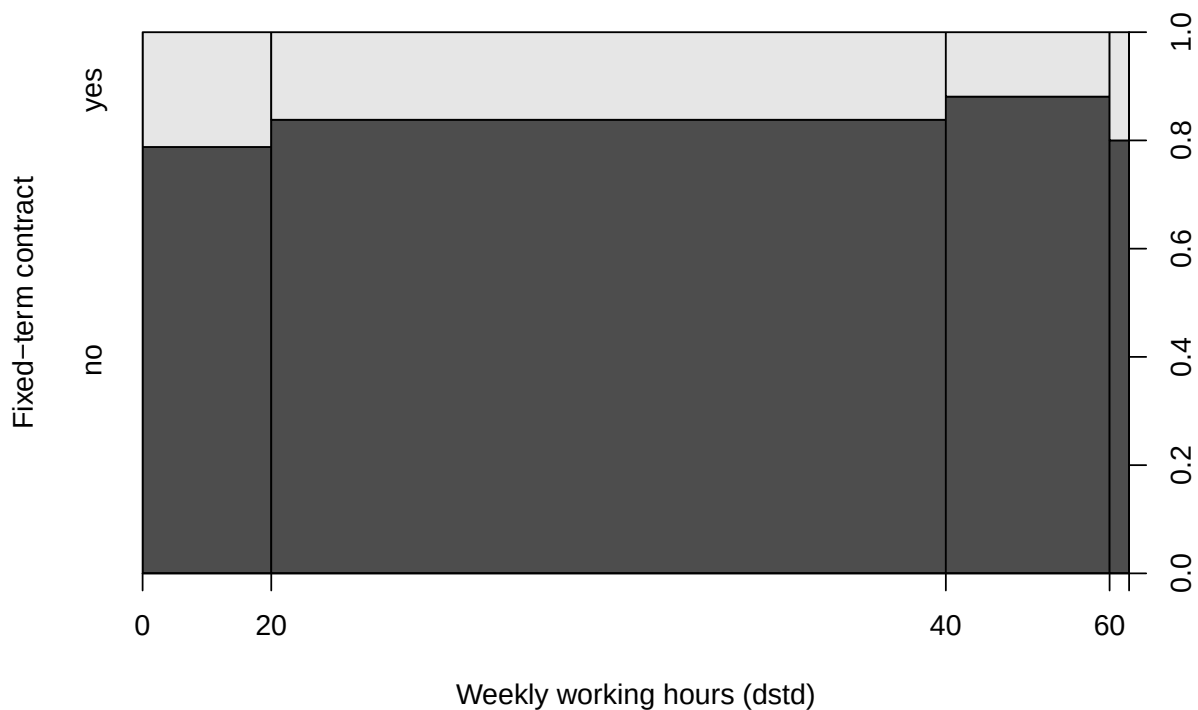


Figure 6: Relationship between Weekly Working Hours and Fixed-Term Rental Agreements.

The spineplot indicates only a weak association between weekly working hours and rental contract type. In all working-hour groups, permanent rental contracts clearly predominate over fixed-term contracts. The proportion of fixed-term contracts appears to be somewhat higher among respondents with lower weekly working hours (4-20) and smaller in the middle range (20-40), whereas the pattern at very high working hours is less clear due to the small number of observations. Overall, the visual evidence suggests only limited differences in rental contract type across working-hour groups.

3.2.2 kjahr (Years Since Education) vs. wfrist (Fixed-Term Agreement)

```
# Spinogram for second metric predictor
spineplot(
  mc_analysis$wfrist ~ mc_analysis$kjahr,
  xlab = "Years since education completion (kjahr)",
  ylab = "Fixed-term contract",
  breaks = 5
)
```

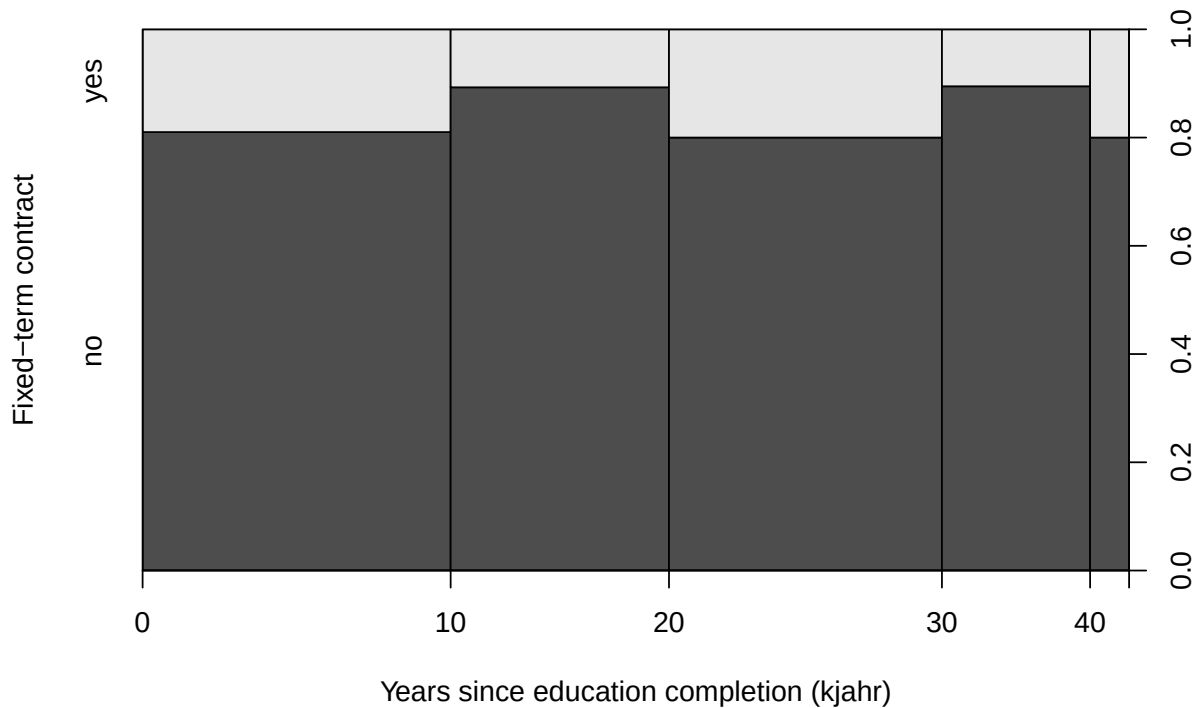


Figure 7: Relationship between Career Stage (Years Since Education) and Fixed-Term Rental Agreements.

The spineplot indicates only a weak association between years since education completion and rental contract type. In all groups, permanent rental contracts clearly predominate over fixed-term contracts. Although the proportion of fixed-term agreements varies slightly across the categories, no clear or consistent trend can be observed. Overall, the visual evidence suggests that time since educational completion is only weakly related to rental contract type in the analyzed sample.

3.2.3 bfst (Family Status) vs. wfrist (Fixed-Term Agreement)

```
# 2D contingency table
bfst_wfrist <- table(mc_analysis$bfst, mc_analysis$wfrist)
cap_bfst_wfrist <- sprintf(
  "Cross-tabulation of Family Status and Fixed-Term Rental Agreements. Single (fixed-term %d
  bfst_wfrist[1, 1],
  bfst_wfrist[1, 2],
  bfst_wfrist[2, 1],
  bfst_wfrist[2, 2],
  bfst_wfrist[3, 1],
  bfst_wfrist[3, 2],
  sum(bfst_wfrist)
)
pander(bfst_wfrist, caption = cap_bfst_wfrist)
```

Table 16: Cross-tabulation of Family Status and Fixed-Term Rental Agreements. Single (fixed-term 20, permanent 116), married (11, 61), widowed/divorced (10, 35). N = 253.

	yes	no
single	20	116
married	11	61
widowed/divorced	10	35

```
# Mosaic plot
mosaicplot(
  bfst_wfrist,
  xlab = "Family status",
  ylab = "Fixed-term contract",
  color = TRUE
)
```

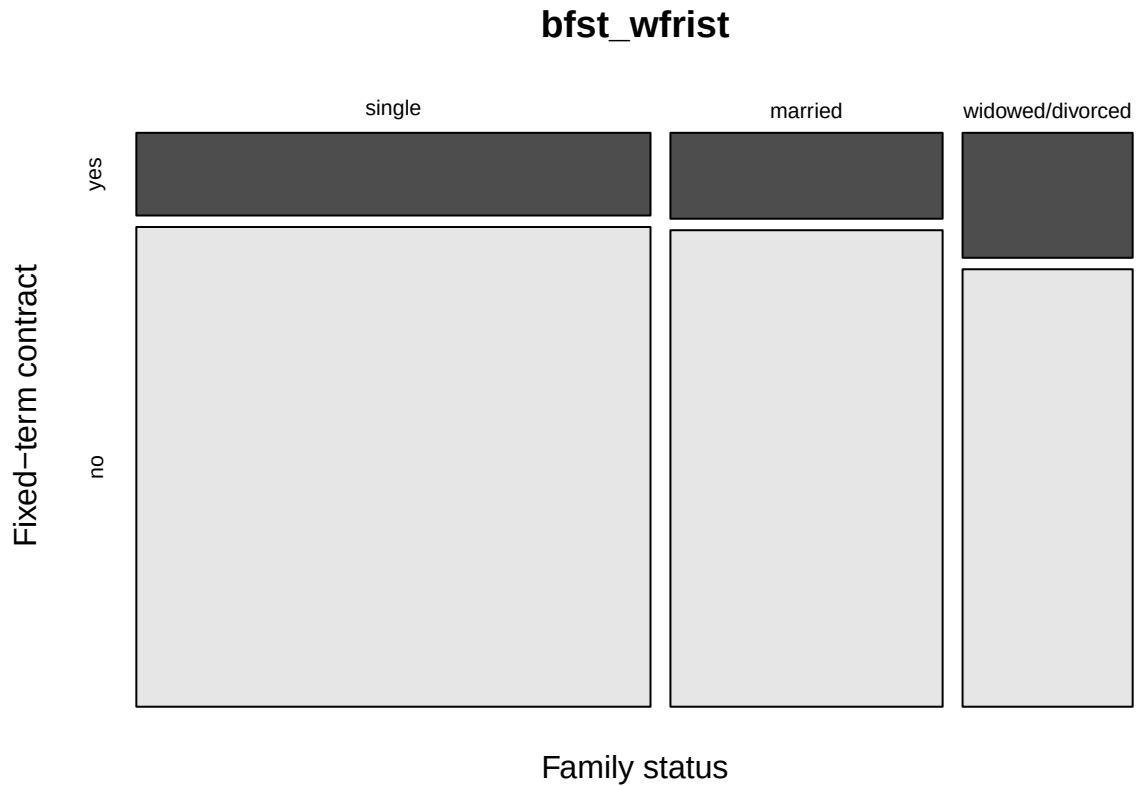


Figure 8: Fixed-Term Agreement by Family Status.

The mosaic plot shows that permanent rental contracts ($wfrist = no$) dominate in all family-status groups. However, their proportion appears to be highest among single respondents, somewhat lower among married respondents and lowest among widowed/divorced respondents. Conversely, the share of fixed-term contracts seems to be largest in the widowed/divorced group. This suggests some differences in rental contract type by family status, although the overall association does not appear to be very strong.

3.2.4 xeinw (Community size) vs. wfrist (Fixed-Term Agreement)

```
# 2D contingency table
xeinw_wfrist <- table(mc_analysis$xeinw, mc_analysis$wfrist)
cap_xeinw_wfrist <- sprintf(
  "Cross-tabulation of Community Size and Fixed-Term Rental Agreements. Small (fixed-term %d",
  xeinw_wfrist[1, 1],
  xeinw_wfrist[1, 2],
  xeinw_wfrist[2, 1],
```

```

xeinw_wfrist[2, 2],
xeinw_wfrist[3, 1],
xeinw_wfrist[3, 2],
xeinw_wfrist[4, 1],
xeinw_wfrist[4, 2],
sum(xeinw_wfrist)
)
pander(xeinw_wfrist, caption = cap_xeinw_wfrist)

```

Table 17: Cross-tabulation of Community Size and Fixed-Term Rental Agreements. Small (fixed-term 5, permanent 16), medium (15, 68), large (15, 103), very large (6, 25). N = 253.

	yes	no
up to 2000	5	16
2001-10000	15	68
10001-100000	15	103
100001+	6	25

```

# Mosaic plot
mosaicplot(
  xeinw_wfrist,
  xlab = "Community size",
  ylab = "Fixed-term contract",
  color = TRUE
)

```

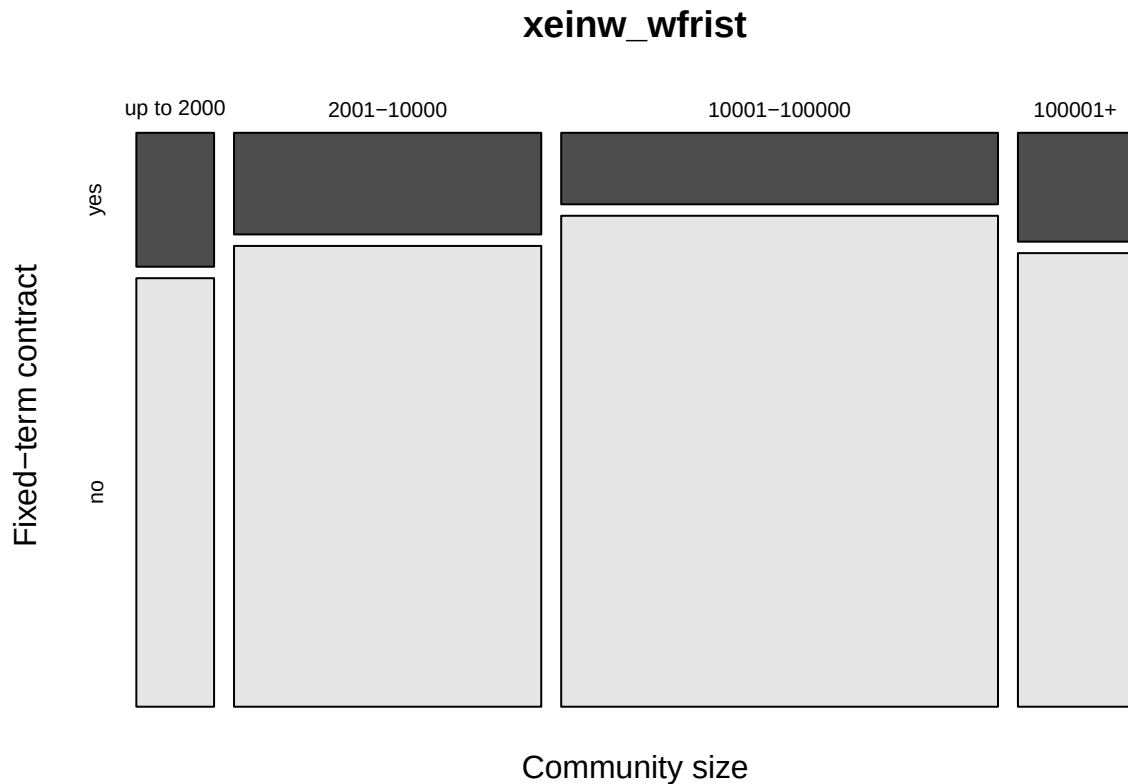


Figure 9: Fixed-Term Agreement by Community Size.

The mosaic plot shows that permanent rental contracts (*wfrist* = *no*) dominate in all community-size groups. Their proportion appears to be highest in communities with 10001 – 100000 inhabitants and somewhat lower in the other categories, especially in the smallest communities. Accordingly, fixed-term rental contracts seem to be slightly more common in smaller communities. However, the differences are not very large, so the overall relationship between community size and rental contract type appears rather weak.

3.3 Contingency Tables: Joint Categorical Influences

3.3.1 3D Contingency Table: *bfst* × *xeinw* × *wfrist*

This three-way table is essential for logistic regression with multiple categorical predictors, showing the joint effect of family status and community size on fixed-term contract status:

```
# 3D contingency table
contingency_3d <- ftable(
  mc_analysis$bfst,
```

```

mc_analysis$xeinw,
mc_analysis$wfrist
)
cap_3d <- "Three-way Cross-tabulation of Family Status, Community Size and Fixed-Term Rental Agreements (N = 253). Essential for logistic regression with multiple categorical predictors. Permanent contracts predominate across all family status and community size combinations."
pander(contingency_3d, caption = cap_3d)

```

Table 18: Three-way Cross-tabulation of Family Status, Community Size and Fixed-Term Rental Agreements (N = 253). Essential for logistic regression with multiple categorical predictors. Permanent contracts predominate across all family status and community size combinations.

		yes	no
single	up to 2000	1	7
	2001-10000	10	47
	10001-100000	5	49
	100001+	4	13
married	up to 2000	4	7
	2001-10000	0	11
	10001-100000	6	36
	100001+	1	7
widowed/divorced	up to 2000	0	2
	2001-10000	5	10
	10001-100000	4	18
	100001+	1	5

The three-dimensional contingency table suggests that the relationship between family status, community size and rental contract type is not very strong overall, but some differences can still be observed.

Across nearly all combinations of family status and community size, permanent rental contracts are more common than fixed-term contracts. At the same time, the share of fixed-term contracts appears to vary somewhat between subgroups. Among single respondents, fixed-term contracts seem to occur somewhat more often in smaller communities and in the largest communities than in medium-sized communities.

Among married respondents, permanent contracts dominate particularly strongly, especially in communities with 2001 – 10000 inhabitants, where no fixed-term contracts are observed. In the widowed/divorced group, fixed-term contracts appear somewhat more common in communities with 2001 – 10000 inhabitants than in the other community-size categories.

Overall, the table points to some subgroup differences.

3.4 Interaction Analyses: Pairwise Predictor Effects

3.4.1 Interaction 1: $dstd \times bfst$ on $wfrist$

```
# Grouped spinogram: Weekly hours by family status (3 categories now)
par(mfrow = c(2, 2))
for (level in levels(mc_analysis$bfst)) {
  subset_data <- subset(mc_analysis, bfst == level)
  spineplot(
    subset_data$wfrist ~ subset_data$dstd,
    main = paste("Family Status:", level),
    xlab = "Weekly working hours",
    ylab = "Fixed-term contract",
    breaks = 4,
    ylim = c(0, 1)
  )
}
par(mfrow = c(1, 1))
```

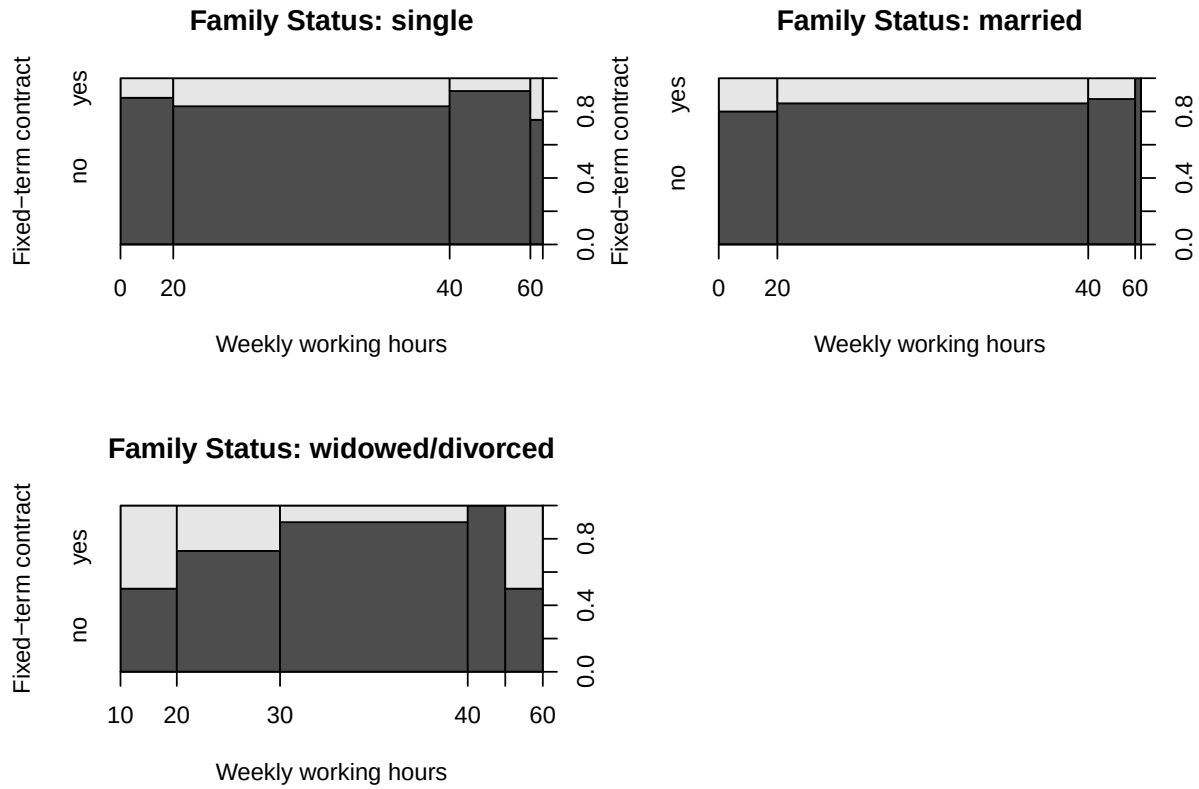


Figure 10: Interaction between Weekly Working Hours and Family Status on Fixed-Term Rental Agreements. Three grouped spinograms (1×3 layout) for each family status category (N = 253).

The grouped spinograms suggest that the relationship between weekly working hours and rental contract type differs only slightly by family status. In all three family-status groups, permanent rental contracts dominate across most working-hour ranges.

For single and married respondents, the share of fixed-term contracts appears relatively small and fairly stable across the observed working-hour groups, indicating no strong interaction between working hours and family status.

In the widowed/divorced group, the proportion of fixed-term contracts seems somewhat higher at lower working-hour levels and lower at higher working-hour levels.

Overall, the plots do not point to a strong interaction effect between weekly working hours and family status on rental contract type.

3.4.2 Interaction 2: dstd × xeinw on wfrist

```
# Grouped spinogram: Weekly hours by community size (4 categories)
par(mfrow = c(2, 2))
for (level in levels(mc_analysis$xeinw)) {
  subset_data <- subset(mc_analysis, xeinw == level)
  spineplot(
    subset_data$wfrist ~ subset_data$dstd,
    main = paste("Community size:", level),
    xlab = "Weekly working hours",
    ylab = "Fixed-term contract",
    breaks = 4,
    ylim = c(0, 1)
  )
}
```



Figure 11: Interaction between Weekly Working Hours and Community Size on Fixed-Term Rental Agreements. Four grouped spinograms (2×2 layout) for each settlement size category (N = 253).


```
par(mfrow = c(1, 1))
```

The grouped spinograms suggest that the relationship between weekly working hours and rental contract type varies only slightly across community-size groups. In all four community-size categories, permanent rental contracts clearly dominate over fixed-term contracts across most working-hour ranges. In the smallest communities, the distribution appears somewhat more irregular, with a comparatively larger share of fixed-term contracts in some working-hour ranges. Overall, the plots do not indicate a strong interaction between weekly working hours and community size in explaining rental contract type.

3.4.3 Interaction 3: $kjahr \times bfst$ on $wfrist$

```
# Grouped spinogram: Years since education by family status (3 categories now)
par(mfrow = c(2, 2))
for (level in levels(mc_analysis$bfst)) {
  subset_data <- subset(mc_analysis, bfst == level)
  spineplot(
    subset_data$wfrist ~ subset_data$kjahr,
    main = paste("Family Status:", level),
    xlab = "Years since education completion",
    ylab = "Fixed-term contract",
    breaks = 5,
    ylim = c(0, 1)
  )
}
par(mfrow = c(1, 1))
```

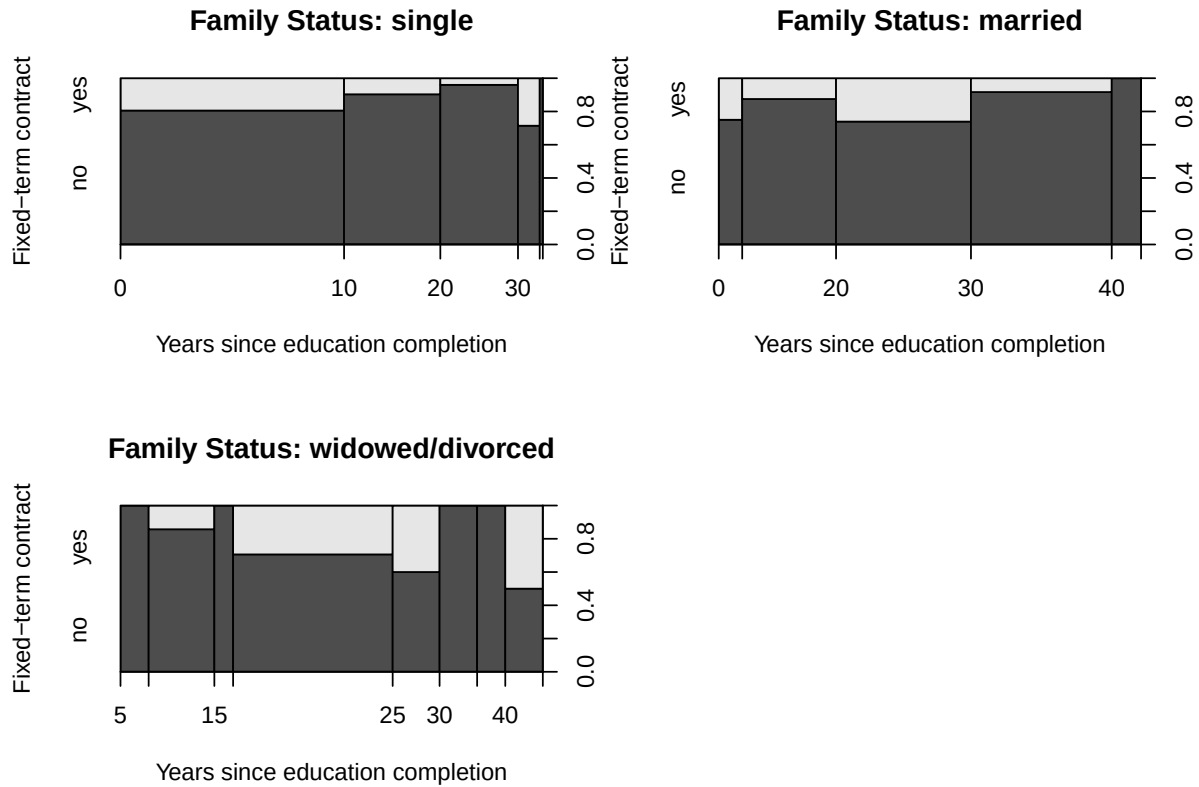


Figure 12: Interaction between Career Stage (Years Since Education) and Family Status on Fixed-Term Rental Agreements. Three grouped spinograms for each family status category (N = 253).

The grouped spinograms indicate that the relationship between years since education completion and rental contract type differs only slightly across family-status groups. In all groups, permanent rental contracts predominate over fixed-term contracts throughout most of the observed range. For single and married respondents, the proportion of fixed-term contracts appears relatively small and fairly stable across the categories of years since education completion, suggesting no strong interaction between these two variables. In the widowed/divorced group, the pattern appears somewhat more irregular, with greater fluctuation in the share of fixed-term contracts across the range of years since education completion. Overall, the plots do not suggest a strong interaction effect between family status and years since education completion in explaining rental contract type.

3.4.4 Interaction 4: $kjahr \times xeinw$ on $wfrist$

```
# Grouped spinogram: Years since education by community size
par(mfrow = c(2, 2))
for (level in levels(mc_analysis$xeinw)) {
  subset_data <- subset(mc_analysis, xeinw == level)
  spineplot(
    subset_data$wfrist ~ subset_data$kjahr,
    main = paste("Community size:", level),
    xlab = "Years since education completion",
    ylab = "Fixed-term contract",
    breaks = 5,
    ylim = c(0, 1)
  )
}
```

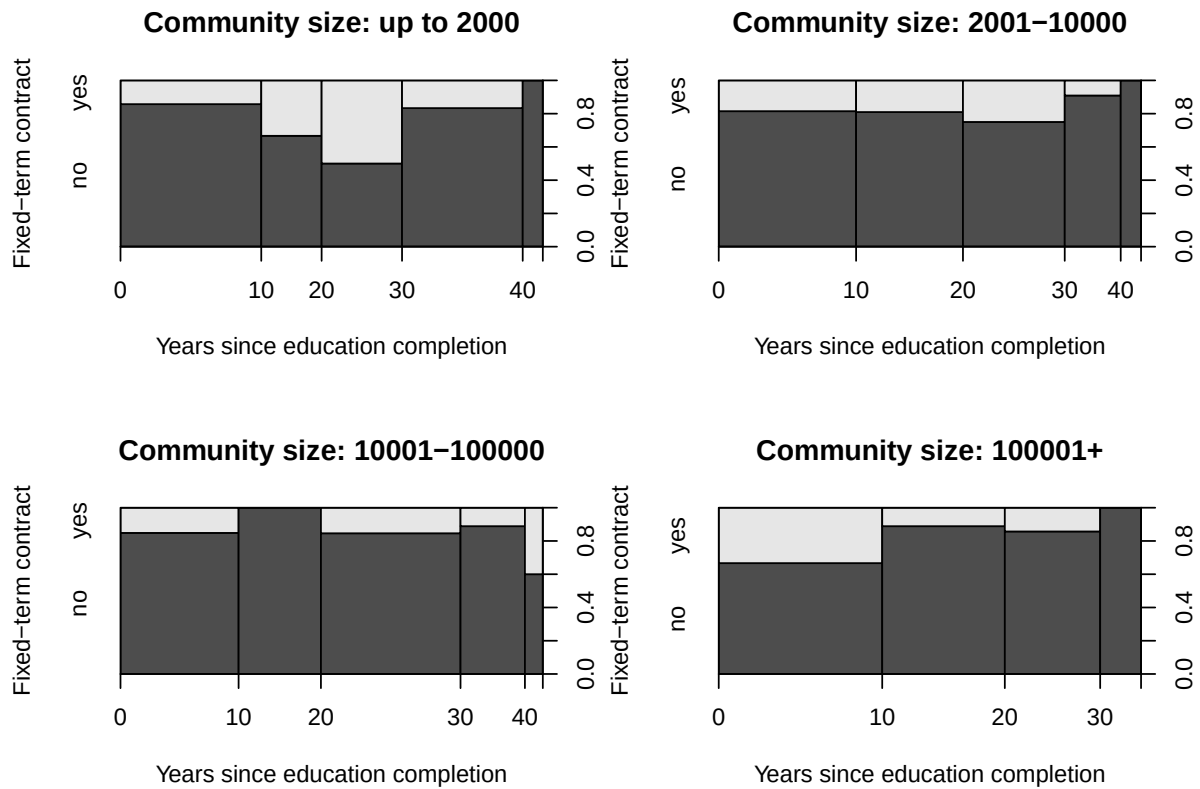


Figure 13: Interaction between Career Stage (Years Since Education) and Community Size on Fixed-Term Rental Agreements. Four grouped spinograms for each settlement size category (N = 253).

```
par(mfrow = c(1, 1))
```

The grouped spinograms suggest that the relationship between years since education completion and rental contract type varies only slightly across community-size groups. In all community-size categories, permanent rental contracts clearly predominate over fixed-term contracts across most of the observed range. In the smallest communities, the distribution appears somewhat more irregular, with stronger fluctuations across the categories of years since education completion. Overall, the plots do not provide evidence of a strong interaction between community size and years since education completion in explaining rental contract type.

3.4.5 Interaction 5: $\text{bfst} \times \text{xeinw}$ on wfrist (Doubledecker Plot)

```
mc_plot <- mc_analysis
levels(mc_plot$bfst) <- c("single", "married", "wid/div")
levels(mc_plot$xeinw) <- c("S", "M", "L", "VL") # S = up to 2000, M = 2001-10000, L = 10001-10000, VL = 10001-100000

doubledecker(
  wfrist ~ bfst + xeinw,
  data = mc_plot,
  gp = gpar(fill = c("grey85", "grey40")),
  gp_text = gpar(fontsize = 9)
)
```

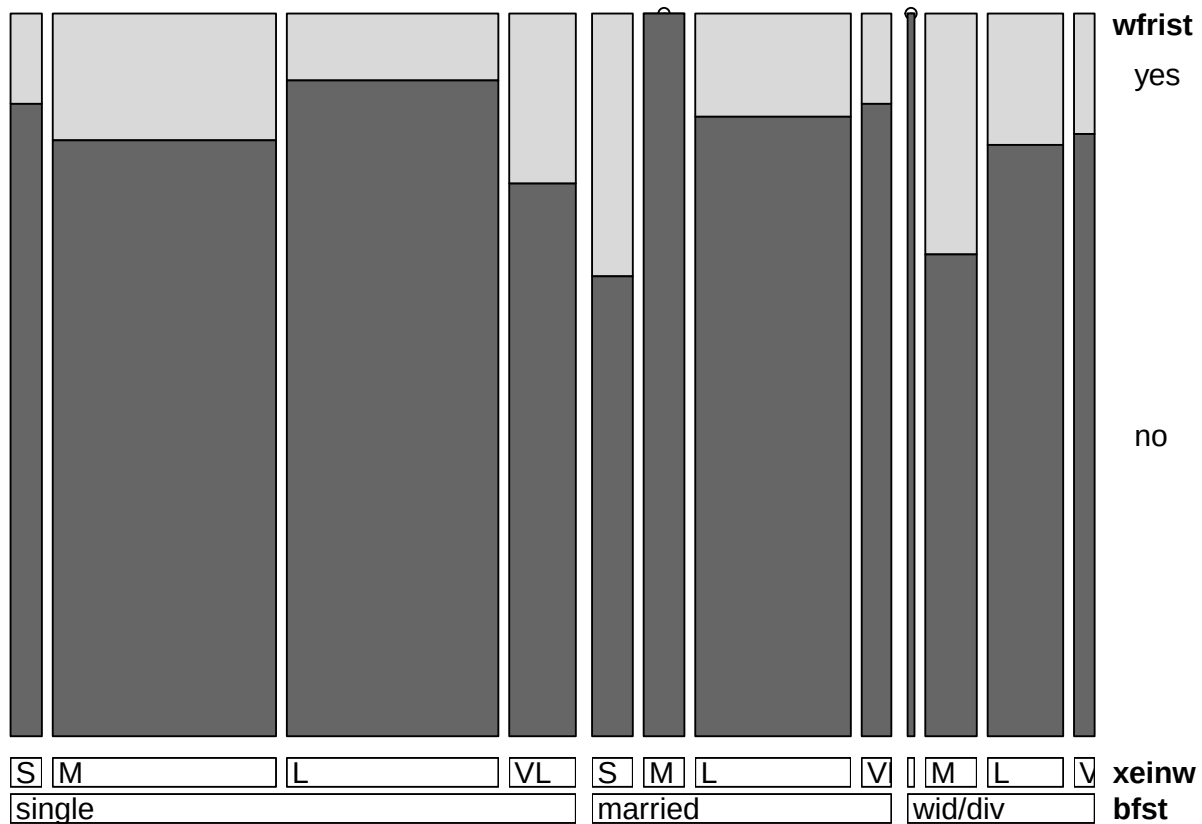


Figure 14: Joint Relationship between Family Status, Community Size and Fixed-Term Rental Agreements (Doubled-decker Plot). Simultaneous display of categorical interaction ($N = 253$).

The doubled-decker plot illustrates the joint relationship of family status and community size with rental contract type. Across almost all combinations of family status and community size, permanent rental contracts ($wfrist = no$) clearly dominate over fixed-term contracts ($wfrist = yes$). The largest subgroup in the sample consists of single respondents, especially those living in medium-sized communities (2001 – 10000). Within each family-status group, the proportion of fixed-term contracts varies somewhat across community-size categories, but these differences are generally modest. Some more noticeable fluctuations appear in the married and widowed/divorced groups, particularly in the smallest subgroups. Overall, the plot suggests only a weak joint effect of family status and community size on rental contract type.

3.5 Contingency Table Analysis for Categorical Predictors

For categorical predictors in logistic regression, contingency tables provide frequency information and allow visual inspection of the distribution of observations across categories:

3.5.1 Family Status × Fixed-Term Contract Type

```
bfst_wfrist_tab <- table(
  Family_Status = mc_analysis$bfst,
  Fixed_Term_Contract = mc_analysis$wfrist
)
pander(
  bfst_wfrist_tab,
  caption = "Contingency Table: Family Status × Fixed-Term Contract Type"
)
```

Table 19: Contingency Table: Family Status × Fixed-Term Contract Type

	yes	no
single	20	116
married	11	61
widowed/divorced	10	35

```
# Row proportions (conditional distribution within each family status)
pander(
  prop.table(bfst_wfrist_tab, margin = 1),
  caption = "Row Proportions: Proportion of Fixed-Term Contracts by Family Status"
)
```

Table 20: Row Proportions: Proportion of Fixed-Term Contracts by Family Status

	yes	no
single	0.1471	0.8529
married	0.1528	0.8472
widowed/divorced	0.2222	0.7778

3.5.2 Community Size × Fixed-Term Contract Type

```
xeinw_wfrist_tab <- table(
  Community_Size = mc_analysis$xeinw,
  Fixed_Term_Contract = mc_analysis$wfrist
)
```

```
pander(
  xeinw_wfrist_tab,
  caption = "Contingency Table: Community Size × Fixed-Term Contract Type"
)
```

Table 21: Contingency Table: Community Size × Fixed-Term Contract Type

	yes	no
up to 2000	5	16
2001-10000	15	68
10001-100000	15	103
100001+	6	25

```
# Row proportions (conditional distribution within each community-size group)
pander(
  prop.table(xeinw_wfrist_tab, margin = 1),
  caption = "Row Proportions: Proportion of Fixed-Term Contracts by Community Size"
)
```

Table 22: Row Proportions: Proportion of Fixed-Term Contracts by Community Size

	yes	no
up to 2000	0.2381	0.7619
2001-10000	0.1807	0.8193
10001-100000	0.1271	0.8729
100001+	0.1935	0.8065

3.5.3 Family Status × Community Size (Joint Distribution)

```
bfst_xeinw_tab <- table(
  Family_Status = mc_analysis$bfst,
  Community_Size = mc_analysis$xeinw
)
pander(
  bfst_xeinw_tab,
  caption = "Contingency Table: Family Status × Community Size"
)
```

Table 23: Contingency Table: Family Status $\times$ Community Size

	up to 2000	2001-10000	10001-100000	100001+
single	8	57	54	17
married	11	11	42	8
widowed/divorced	2	15	22	6

These contingency tables provide cell counts that allow verification of sufficient observations in every combination of categorical predictor levels, an important prerequisite for reliable inference in logistic regression with categorical variables.

3.6 Summary and Transition to Analysis

Overall, the descriptive analysis shows that permanent rental agreements clearly predominate in the analyzed sample of renters from Southern Austria, while fixed-term contracts represent a much smaller share. Across the selected predictors, the bivariate patterns suggest that family status and community size show somewhat clearer differences in contract type than the two metric predictors.

For weekly working hours and years since education completion, the visualizations point to only weak relationships with rental contract type, as permanent agreements dominate in nearly all groups and no strong monotonic trends are visible. The joint displays of predictors also do not indicate pronounced interaction effects. While some subgroup differences appear in the grouped spinograms and doubledecker plot, these patterns are generally small and in some cases based on rather few observations.

Taken together, the descriptive findings suggest that fixed-term rental contracts may vary somewhat by respondents family situation and settlement context, while labor-related and life-course factors appear less strongly associated at the descriptive level. These results motivate the next step of the analysis, namely to examine whether the observed differences remain meaningful at the population level when the predictors are considered simultaneously in a binary logistic regression model.

4 Regression Modeling

4.1 Naive Model and Diagnostics

We begin by fitting a naive logistic regression model that includes all main effects of the four predictors without interaction terms:


```
# Fit naive model with all main effects
model_naive <- glm(
  wfrist ~ dstd + kjahr + bfst + xeinw,
  family = binomial(link = "logit"),
  data = mc_analysis
)

# Summary of the naive model
pander(
  summary(model_naive),
  caption = "Naive Model: Binary Logistic Regression with Main Effects Only"
)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.6563	0.7931	0.8274	0.408
dstd	0.01096	0.01503	0.7292	0.4659
kjahr	0.02013	0.01852	1.087	0.2769
bfstmarried	-0.3244	0.5076	-0.6391	0.5227
bfstwidowed/divorced	-0.7516	0.4902	-1.533	0.1252
xeinw2001-10000	0.3127	0.6163	0.5073	0.6119
xeinw10001-100000	0.7351	0.5981	1.229	0.219
xeinw100001+	0.3166	0.7032	0.4503	0.6525

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	224.2 on 252 degrees of freedom
Residual deviance:	218.5 on 245 degrees of freedom

In the naive logistic regression model, none of the included predictors are statistically significant at the 5% level. The model has a residual deviance of 218.46 compared to a null deviance of 224.19, indicating only a small improvement over the intercept-only model. The AIC is 234.46. Overall, the model provides limited evidence that *dstd*, *kjahr*, *bfst* or *xeinw* are strong predictors of *wfrist*. Given the small number of observations in several subgroups, these findings should be interpreted cautiously, as sparse data may reduce statistical power and produce unstable estimates.

However, the naive model provides a baseline for comparison. To assess the model fit and check for potential problems (outliers, leverage, nonlinearity), we perform diagnostic analyses using the DHARMa package, which is recommended for binary logistic regression:

```
# Generate DHARMA residuals for binomial model
simulationOutput <- simulateResiduals(fittedModel = model_naive, n = 1000)

# Plot diagnostics
plot(simulationOutput)
```

DHARMA residual

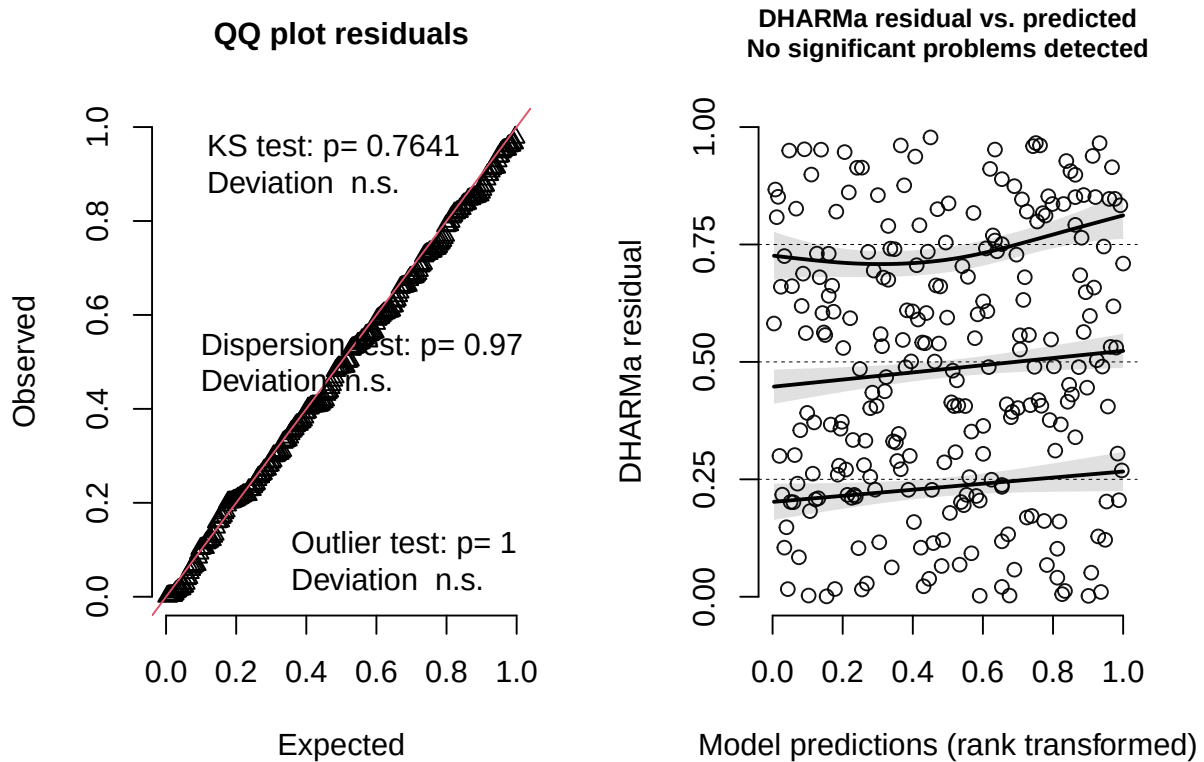


Figure 15: DHARMA Simulation-Based Diagnostic Plots for Naive Model. Includes residual uniformity test, dispersion test, and outlier identification.

The plot of DHARMA residuals against predicted values also shows no significant problems. The residuals are spread fairly evenly across the range of predicted values, and DHARMA reports no significant pattern. This suggests that there is no clear evidence of model misspecification based on the residual diagnostics.

Additionally, we examine influential observations and potential outliers:

```

# Calculate influence measures
influence_measures <- cooks.distance(model_naive)

# Identify potentially influential observations
influential_threshold <- 4 / nrow(mc_analysis)
influential_obs <- which(influence_measures > influential_threshold)

if (length(influential_obs) > 0) {
  pander(
    mc_analysis[influential_obs, ],
    caption = paste(
      "Observations with high influence on model estimates, with",
      length(influential_obs),
      "influential observations (Cook's D >",
      round(influential_threshold, 4),
      ")"
    )
  )
}

```

Table 24: Observations with high influence on model estimates, with 23 influential observations (Cook's D > 0.0158)

	wfrist	dstd	kjahr	bfst	xeinw
1	yes	30	26	widowed/divorced	100001+
482	yes	52	6	single	100001+
791	yes	72	7	single	100001+
2002	yes	6	6	single	100001+
2675	yes	30	9	single	100001+
3553	yes	10	30	married	10001-100000
3665	yes	60	44	widowed/divorced	10001-100000
3697	yes	25	12	married	100001+
5743	yes	38.5	35	married	10001-100000
5917	yes	40	27	widowed/divorced	10001-100000
6343	yes	40	12	widowed/divorced	2001-10000
6584	yes	40	32	single	up to 2000
6788	yes	40	14	married	up to 2000
6789	yes	32	10	married	up to 2000
6792	yes	45	21	married	up to 2000
6793	yes	18	22	married	up to 2000
6938	yes	20	23	widowed/divorced	2001-10000
6939	yes	40	34	single	2001-10000

	wfrist	dstd	kjahr	bfst	xeinw
6986	yes	20	23	widowed/divorced	10001-100000
6987	yes	55	42	widowed/divorced	10001-100000
6988	yes	38.5	0	single	10001-100000
7175	yes	18	21	widowed/divorced	2001-10000
7616	yes	40	39	married	10001-100000

The influence analysis identified 23 observations with Cook’s distance values above the chosen threshold of 0.0158. This suggests that these observations have a relatively stronger influence on the estimated model coefficients compared with the rest of the data. Interestingly, all influential observations correspond to cases where the outcome wfrist is “yes”. This indicates that the model estimates may be especially sensitive to the relatively small number of “yes” cases in the dataset.

4.2 Model Building and Variable Selection

To identify the most important predictors and whether interaction terms improve the model, we perform model selection using multiple procedures: backward elimination, forward selection and stepwise selection.

```
# Full model with all main effects and two-way interactions
model_full <- glm(
  wfrist ~ (dstd + kjahr + bfst + xeinw)^2,
  family = binomial(link = "logit"),
  data = mc_analysis
)

pander(
  summary(model_full),
  caption = "Full Model: Binary Logistic Regression with All Main Effects and Two-Way Interactions"
)
```

Table 25: Table continues below

	Estimate	Std. Error
(Intercept)	4.062	3.346
dstd	-0.07558	0.07072
kjahr	0.1063	0.1128
bfstmarried	-5.033	2.503

	Estimate	Std. Error
bfstwidowed/divorced	14.4	2783
xeinw2001-10000	-4.891	3.347
xeinw10001-100000	-1.4	3.07
xeinw100001+	-3.376	3.337
dstd:kjahr	-0.0009184	0.001938
dstd:bfstmarried	0.07187	0.04989
dstd:bfstwidowed/divorced	0.04058	0.05181
dstd:xeinw2001-10000	0.1225	0.07165
dstd:xeinw10001-100000	0.06607	0.06367
dstd:xeinw100001+	0.07209	0.06898
kjahr:bfstmarried	-0.01452	0.06312
kjahr:bfstwidowed/divorced	-0.05197	0.05414
kjahr:xeinw2001-10000	-0.01535	0.08685
kjahr:xeinw10001-100000	-0.06926	0.07448
kjahr:xeinw100001+	0.006555	0.09642
bfstmarried:xeinw2001-10000	18.18	1135
bfstwidowed/divorced:xeinw2001-10000	-15.86	2783
bfstmarried:xeinw10001-100000	2.284	1.948
bfstwidowed/divorced:xeinw10001-100000	-15.37	2783
bfstmarried:xeinw100001+	2.671	2.419
bfstwidowed/divorced:xeinw100001+	-15.28	2783

	z value	Pr(> z)
(Intercept)	1.214	0.2247
dstd	-1.069	0.2852
kjahr	0.9422	0.3461
bfstmarried	-2.01	0.04438
bfstwidowed/divorced	0.005172	0.9959
xeinw2001-10000	-1.461	0.1439
xeinw10001-100000	-0.4561	0.6483
xeinw100001+	-1.012	0.3117
dstd:kjahr	-0.474	0.6355
dstd:bfstmarried	1.44	0.1498
dstd:bfstwidowed/divorced	0.7833	0.4335
dstd:xeinw2001-10000	1.71	0.08725
dstd:xeinw10001-100000	1.038	0.2994
dstd:xeinw100001+	1.045	0.296
kjahr:bfstmarried	-0.2301	0.818
kjahr:bfstwidowed/divorced	-0.9599	0.3371

	z value	Pr(> z)
kjahr:xeinw2001-10000	-0.1767	0.8597
kjahr:xeinw10001-100000	-0.9298	0.3525
kjahr:xeinw100001+	0.06799	0.9458
bfstmarried:xeinw2001-10000	0.01602	0.9872
bfstwidowed/divorced:xeinw2001-10000	-0.005697	0.9955
bfstmarried:xeinw10001-100000	1.172	0.241
bfstwidowed/divorced:xeinw10001-100000	-0.005522	0.9956
bfstmarried:xeinw100001+	1.104	0.2696
bfstwidowed/divorced:xeinw100001+	-0.00549	0.9956

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	224.2 on 252 degrees of freedom
Residual deviance:	200.2 on 228 degrees of freedom

4.2.1 Backward Selection

```
model_backward <- step(model_full, direction = "backward", trace = FALSE)

pander(
  summary(model_backward),
  caption = "Backward Selection: Final Model After Stepwise Elimination of Terms"
)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.643	0.1706	9.63	5.954e-22

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	224.2 on 252 degrees of freedom
Residual deviance:	224.2 on 252 degrees of freedom

4.2.2 Forward Selection

```

model_forward <- step(
  model_naive,
  scope = list(lower = model_naive, upper = model_full),
  direction = "forward",
  trace = FALSE
)
pander(
  summary(model_forward),
  caption = "Forward Selection: Final Model After Stepwise Addition of Terms"
)

```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.6563	0.7931	0.8274	0.408
dstd	0.01096	0.01503	0.7292	0.4659
kjahr	0.02013	0.01852	1.087	0.2769
bfstmarried	-0.3244	0.5076	-0.6391	0.5227
bfstwidowed/divorced	-0.7516	0.4902	-1.533	0.1252
xeinw2001-10000	0.3127	0.6163	0.5073	0.6119
xeinw10001-100000	0.7351	0.5981	1.229	0.219
xeinw100001+	0.3166	0.7032	0.4503	0.6525

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	224.2 on 252 degrees of freedom
Residual deviance:	218.5 on 245 degrees of freedom

4.2.3 Stepwise Selection

```

model_stepwise <- step(
  model_naive,
  scope = list(lower = model_naive, upper = model_full),
  direction = "both",
  trace = FALSE
)
pander(
  summary(model_stepwise),
  caption = "Stepwise Selection: Final Model After Bidirectional Stepwise Selection"
)

```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.6563	0.7931	0.8274	0.408
dstd	0.01096	0.01503	0.7292	0.4659
kjahr	0.02013	0.01852	1.087	0.2769
bfstmarried	-0.3244	0.5076	-0.6391	0.5227
bfstwidowed/divorced	-0.7516	0.4902	-1.533	0.1252
xeinw2001-10000	0.3127	0.6163	0.5073	0.6119
xeinw10001-100000	0.7351	0.5981	1.229	0.219
xeinw100001+	0.3166	0.7032	0.4503	0.6525

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	224.2 on 252 degrees of freedom
Residual deviance:	218.5 on 245 degrees of freedom

```
# Compare AIC values across all selection procedures
model_comparison <- data.frame(
  Model = c(
    "Naive (main effects only)",
    "Full (all main effects + 2-way interactions)",
    "Backward",
    "Forward",
    "Stepwise"
  ),
  Formula = c(
    deparse(formula(model_naive)),
    deparse(formula(model_full)),
    deparse(formula(model_backward)),
    deparse(formula(model_forward)),
    deparse(formula(model_stepwise))
  ),
  AIC = c(
    AIC(model_naive),
    AIC(model_full),
    AIC(model_backward),
    AIC(model_forward),
    AIC(model_stepwise)
  )
)

pander(
```



```

model_comparison,
caption = "Model Comparison: AIC values across selection procedures"
)

```

Table 26: Model Comparison: AIC values across selection procedures

Model	Formula	AIC
Naive (main effects only)	wfrist ~ dstd + kjahr + bfst + xeinw	234.5
Full (all main effects + 2-way interactions)	wfrist ~ (dstd + kjahr + bfst + xeinw)^2	250.2
Backward	wfrist ~ 1	226.2
Forward	wfrist ~ dstd + kjahr + bfst + xeinw	234.5
Stepwise	wfrist ~ dstd + kjahr + bfst + xeinw	234.5

```

# Identify the best model
best_model_idx <- which.min(model_comparison$AIC)
best_model_name <- model_comparison$Model[best_model_idx]

if (AIC(model_backward) <= min(AIC(model_forward), AIC(model_stepwise))) {
  model_final <- model_backward
  selection_method <- "Backward"
} else if (AIC(model_forward) <= AIC(model_stepwise)) {
  model_final <- model_forward
  selection_method <- "Forward"
} else {
  model_final <- model_stepwise
  selection_method <- "Stepwise"
}

final_terms <- length(attr(terms(model_final), "term.labels"))

```

The best model based on the AIC is Backward. This result is consistent with the naive logistic regression model, where none of the predictors were statistically significant. It suggests that the available predictors do not provide meaningful explanatory power for predicting wfrist in this dataset.

4.2.4 Drop Term Analysis

To assess the relative importance of each term in the model and the potential impact of removing terms, we examine the change in AIC that would result from removing each term:

```

if (final_terms > 0) {
  pander(
    dropterm(model_final, test = "Chisq"),
    caption = "Drop Term Analysis: Effect of Removing Each Term from the Model"
  )
}

```

Since the final model selected by backward elimination was the intercept-only model, there were no predictors left to evaluate. Therefore, the drop-term analysis was not performed. The value 0 confirms that the final model contains no explanatory variables besides the intercept.

4.3 Final Model Interpretation

4.3.1 Model Summary and Coefficients

```

pander(
  summary(model_final),
  caption = "Final Model Summary: Intercept-Only Model Selected by Backward Elimination"
)

```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.643	0.1706	9.63	5.954e-22

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	224.2 on 252 degrees of freedom
Residual deviance:	224.2 on 252 degrees of freedom

4.3.2 Hypothesis Testing: Likelihood-Ratio Tests

Since backward selection selected the intercept-only model (`wfrist ~ 1`), there are no predictor terms left in the final model. Therefore, likelihood-ratio tests for individual predictors are not applicable. This supports the model selection result that none of the candidate predictors improved model fit sufficiently.

4.3.3 Parameter Significance and Confidence Intervals

```
# Extract coefficients and confidence intervals
final_coefs <- coef(model_final)

# Try to get CIs; use summary if confint fails
try_ci <- try(confint(model_final, level = 0.95), silent = TRUE)

if (inherits(try_ci, "try-error") || !is.matrix(try_ci)) {
  # Fall back to using standard errors from summary
  model_summary <- summary(model_final)
  final_ci <- cbind(
    final_coefs - 1.96 * model_summary$coefficients[, "Std. Error"],
    final_coefs + 1.96 * model_summary$coefficients[, "Std. Error"]
  )
  colnames(final_ci) <- c("2.5 %", "97.5 %")
} else {
  final_ci <- try_ci
}

coef_table <- data.frame(
  Parameter = names(final_coefs),
  Estimate = final_coefs,
  CI_Lower = final_ci[, 1],
  CI_Upper = final_ci[, 2],
  Significant = ifelse(
    (final_ci[, 1] > 0 & final_ci[, 2] > 0) |
    (final_ci[, 1] < 0 & final_ci[, 2] < 0),
    "Yes",
    "No"
  )
)

pander(
  coef_table,
  caption = "Final Model: Parameter Estimates and 95% Confidence Intervals"
)
```

Table 27: Final Model: Parameter Estimates and 95% Confidence Intervals

	Parameter	Estimate	CI_Lower	CI_Upper	Significant
(Intercept)	(Intercept)	1.643	1.309	1.977	Yes

The final model contains only the intercept, with an estimate of 1.643 and a 95% confidence interval from 1.309 to 1.977. This indicates that the overall baseline outcome is statistically significant, but no predictor effects are estimated because all predictors were removed during model selection.

4.3.4 Odds Ratios

```
# Calculate odds ratios and their confidence intervals
odds_ratios <- exp(final_coefs)
odds_ratios_ci <- exp(final_ci)

odds_table <- data.frame(
  Parameter = names(odds_ratios),
  OddsRatio = odds_ratios,
  CI_Lower = odds_ratios_ci[, 1],
  CI_Upper = odds_ratios_ci[, 2]
)

pander(
  odds_table,
  caption = "Final Model: Odds Ratios and 95% Confidence Intervals"
)
```

Table 28: Final Model: Odds Ratios and 95% Confidence Intervals

	Parameter	OddsRatio	CI_Lower	CI_Upper
(Intercept)	(Intercept)	5.171	3.701	7.224

The final model includes only the intercept, with an odds ratio of 5.171 and a 95% confidence interval from 3.701 to 7.224. This shows the overall odds of the outcome in the sample, but it does not represent the effect of any predictor because no predictors were retained in the final model.

4.3.5 Effect Plots

Effect plots provide a visual representation of how predicted probabilities of a fixed-term rental agreement change across the range of predictors:

```
p1 <- plot(ggpredict(model_naive, terms = "dstd")) +  
  labs(  
    title = "Weekly working hours",  
    x = "Weekly working hours",  
    y = "Predicted probability"  
  )  
  
p2 <- plot(ggpredict(model_naive, terms = "kjahr")) +  
  labs(  
    title = "Years since education completion",  
    x = "Years since education completion",  
    y = "Predicted probability"  
  )  
  
p3 <- plot(ggpredict(model_naive, terms = "bfst")) +  
  labs(  
    title = "Family status",  
    x = "Family status",  
    y = "Predicted probability"  
  )  
  
p4 <- plot(ggpredict(model_naive, terms = "xeinw")) +  
  labs(  
    title = "Community size",  
    x = "Community size",  
    y = "Predicted probability"  
  )  
  
(p1 + p2) / (p3 + p4)
```

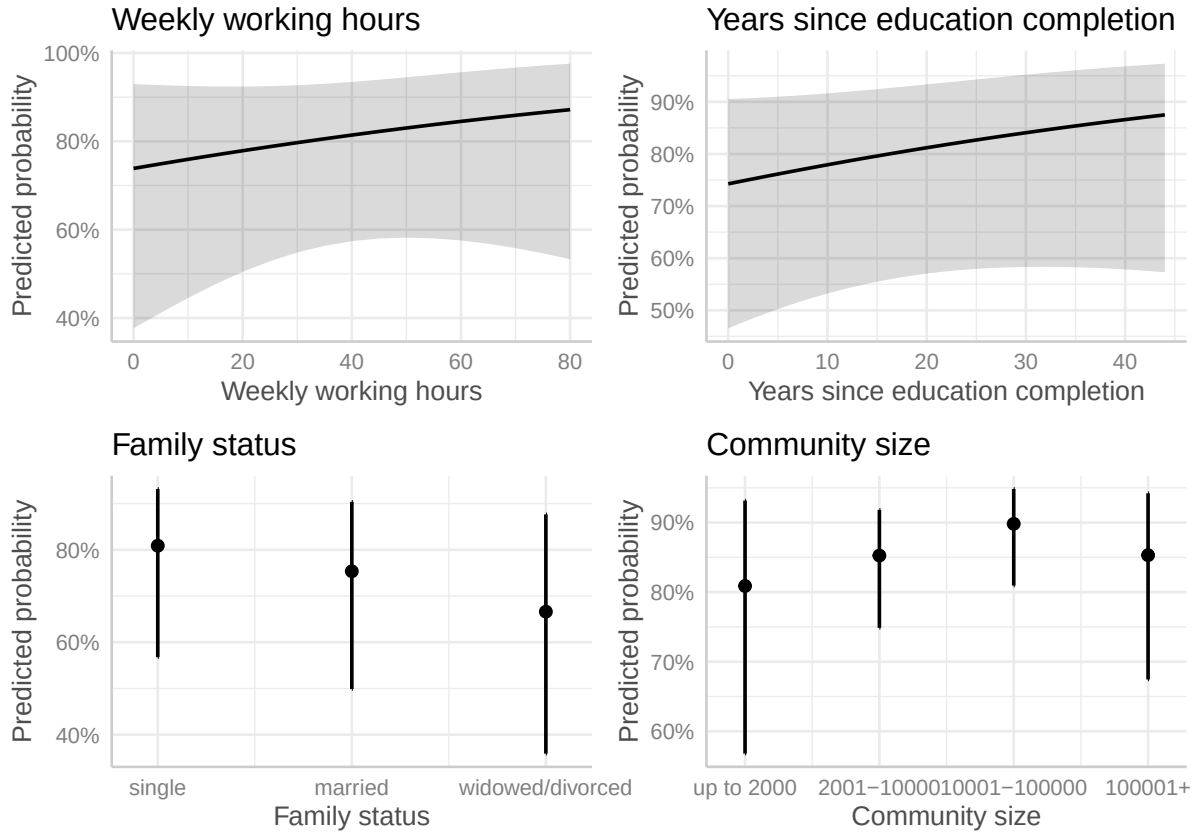


Figure 16: Predicted Probabilities of Fixed-Term Rental Agreements from Naive Model. Four panels showing effects of (a) weekly working hours, (b) years since education completion, (c) family status, and (d) community size on predicted probability. Wide overlapping confidence intervals indicate weak predictor effects.

Since the final selected model was the intercept-only model, effect plots cannot be shown for the final model. Therefore, effect plots from the naive model are presented for descriptive purposes. The plots show that the confidence intervals are wide and mostly overlapping. Therefore, the plots do not provide strong evidence that working hours, years since education completion, family status, or community size clearly affect the probability of the outcome.

4.4 Model Diagnostics for Final Model

The final selected model is an intercept-only model (`wfrist ~ 1`). Therefore, diagnostic plots that assess predictor-specific patterns, such as residual plots for metric predictors or influence plots based on leverage, are not applicable. Since the model contains no explanatory variables, there are no predictor effects or functional forms to diagnose.

However, simulation-based diagnostics can still be used to assess whether the overall model structure is appropriate.

4.4.1 DHARMA Simulation-Based Diagnostics

```
# DHARMA diagnostics for final model
simulationOutput_final <- simulateResiduals(fittedModel = model_final, n = 1000)

plot(simulationOutput_final, main = "DHARMA Diagnostics: Final Model")
```

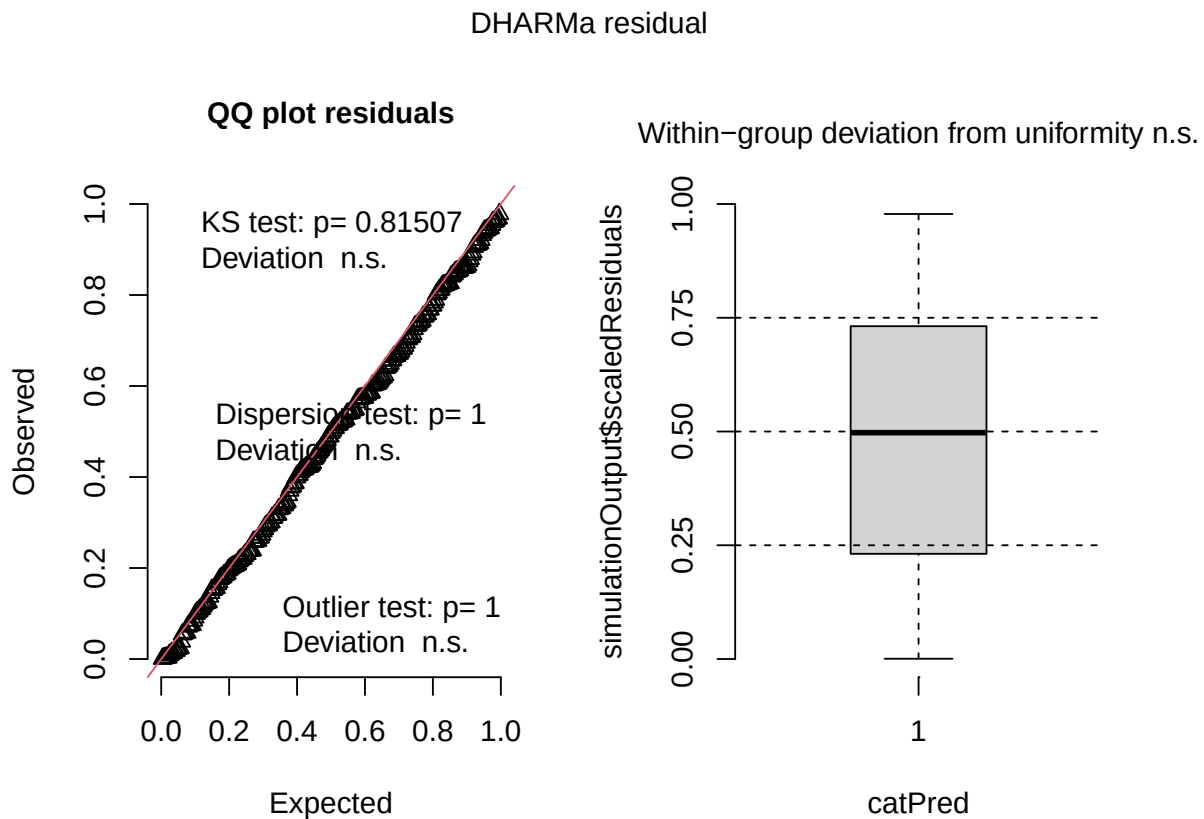


Figure 17: DHARMA Simulation-Based Diagnostic Plots for Final Intercept-Only Model. Assesses overall model fit to the outcome distribution.

The DHARMA diagnostics suggest that the final intercept-only model fits the overall outcome distribution. No significant problems were detected for residual uniformity, dispersion, or outliers. However, because the model contains no predictors, these diagnostics only assess the overall model fit and not any predictor effects.

4.4.2 Posterior Predictive Check

Manual comparison of observed vs. simulated outcomes:

```
# Simulate data from the final model
predictions_final <- predict(
  model_final,
  newdata = mc_analysis,
  type = "response"
)

# Generate simulated binary outcomes
set.seed(42)
simulated_data <- rbinom(
  n = length(predictions_final),
  size = 1,
  prob = predictions_final
)

# Compare observed vs. simulated
obs_fixed_term <- sum(as.numeric(mc_analysis$wfrist) - 1)
sim_fixed_term <- sum(simulated_data)
obs_proportion <- round(obs_fixed_term / length(predictions_final), 3)

# Visualization
set.seed(42)
sims <- replicate(
  1000,
  sum(rbinom(length(predictions_final), 1, predictions_final))
)
mean_sim_fixed_term <- round(mean(sims), 2)
hist(
  sims,
  main = "Posterior Predictive Check",
  xlab = "Number of Fixed-Term Contracts in Simulated Data",
  ylab = "Frequency",
  col = "steelblue"
)
abline(v = obs_fixed_term, col = "red", lwd = 2, label = "Observed")
legend("topright", legend = "Observed", col = "red", lwd = 2)
```

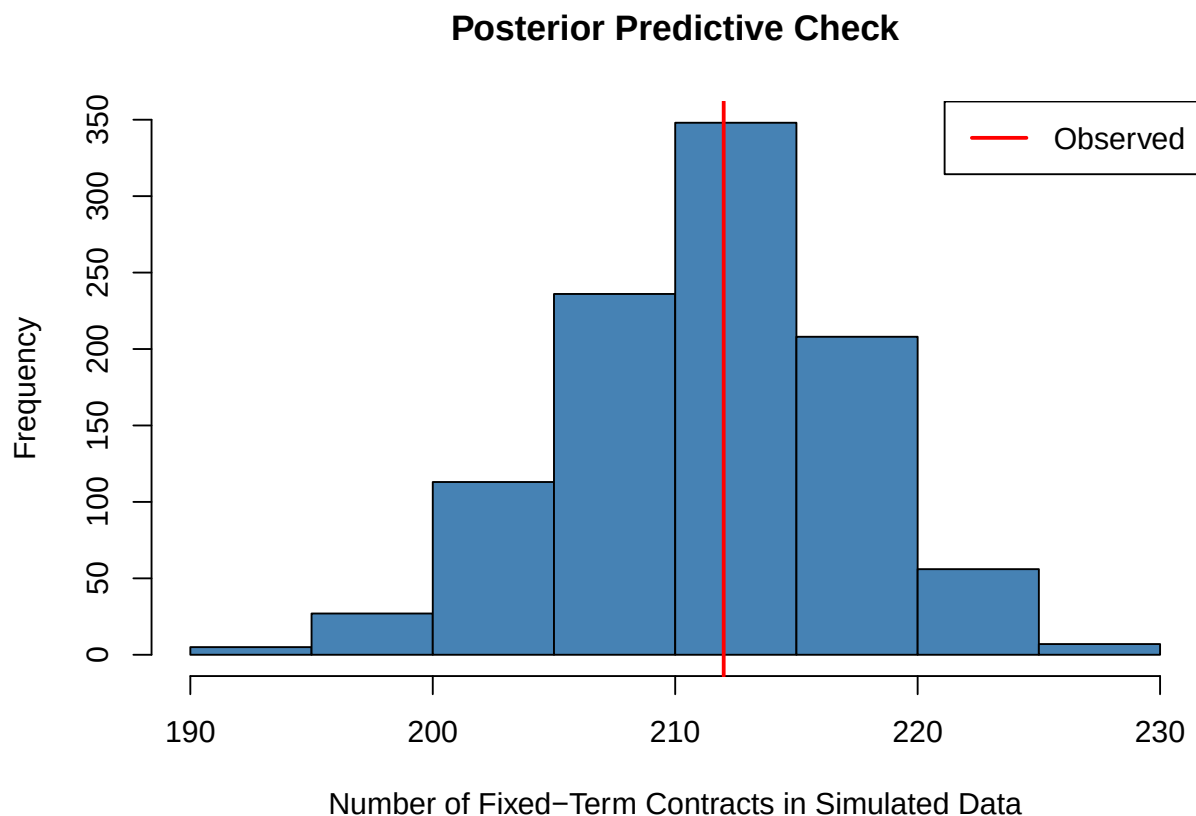



Figure 18: Posterior Predictive Check: Distribution of Simulated Fixed-Term Contract Counts (n=1000 simulations) with Observed Count Marked in Red. Assesses whether the final model generates data consistent with observed outcome frequencies.

The posterior predictive check shows that the observed number of 212 fixed-term contracts falls within the simulated distribution. The mean simulated number of fixed-term contracts across 1000 simulations was 211.8, which closely matches the observed count. The observed proportion of fixed-term contracts in the sample is 83.8%. These values suggest that the intercept-only model captures the overall outcome proportion in the sample reasonably well. However, since the model contains no predictors, this check does not provide evidence about differences between subgroups.

4.5 Relating Results to Research Questions

The fitted logistic regression model addresses the primary research question: **Which factors predict whether a renter in Southern Austria has a fixed-term or permanent rental agreement?**

Based on the model comparison, the final selected model was the intercept-only model. This means that none of the tested predictors - weekly working hours, years since education completion, family status, or community size - improved the prediction of rental contract type enough to be retained in the final model.

Therefore, the results do not provide strong evidence that these variables are meaningful predictors of rental contract type in this sample. The odds ratio of the final model only reflects the overall distribution of the outcome, showing that permanent rental agreements were much more common than fixed-term agreements.

The effect plots from the naive model were included for descriptive purposes. They showed only weak and uncertain patterns, with wide and overlapping confidence intervals. This supports the model selection result that the predictors did not show clear effects.

Overall, the findings do not strongly support the initial expectation that the selected social and regional factors would predict rental contract type. However, the results should be interpreted cautiously because the data were imbalanced and several subgroups contained small numbers of observations.

5 Summary

This report examined whether weekly working hours, years since education completion, family status, and community size predict whether renters in Southern Austria have a fixed-term or permanent rental agreement. The analysis used Austrian microcensus data from 2012 and focused on renters in Carinthia and Styria.

First, the data was cleaned and restricted to complete cases, resulting in a final sample of 253 respondents. Descriptive analyses showed that permanent rental agreements were much more common than fixed-term agreements. Only 41 respondents (16.2%) had a fixed-term contract, while 212 respondents (83.8%) had a permanent contract.

The bivariate and interaction analyses showed only weak relationships between the predictors and rental contract type. Permanent contracts dominated across nearly all groups. Some differences were visible for family status and community size, but these patterns were generally small and often based on sparse subgroup counts.

A binary logistic regression model was then fitted. The naive model included all main effects, but none of the predictors were statistically significant. Different model selection procedures were compared using AIC. The backward selection procedure selected the intercept-only model as the best model, with the lowest AIC. This means that none of the tested predictors improved the prediction of rental contract type enough to remain in the final model.

Therefore, the research question can be answered as follows: in this sample, weekly working hours, years since education completion, family status, and community size do not provide strong evidence for predicting whether a renter has a fixed-term or permanent rental agreement.

The final model mainly reflects the overall distribution of the outcome, where permanent contracts are much more common than fixed-term contracts.

There are several limitations. The outcome variable is imbalanced, with relatively few fixed-term contracts. In addition, several combinations of family status and community size contain very small numbers of observations, including some cells with zero or one fixed-term case. This makes subgroup comparisons less reliable and may reduce the statistical power of the regression models. The influence analysis also showed that several “yes” cases had relatively high influence on the model estimates, suggesting that the results may be sensitive to a small number of observations.

The model diagnostics did not show serious technical problems. DHARMA diagnostics indicated no major issues with residual uniformity, dispersion, or outliers. The posterior predictive check suggested that the final intercept-only model captures the overall outcome proportion reasonably well. However, because the final model contains no predictors, these diagnostics only assess the overall fit and not differences between groups.

The analysis is based only on a subset of renters in Southern Austria from 2012, and the final sample size is relatively small. Therefore, the results should not be interpreted as general evidence for all Austrian renters or for current rental markets.

Further research could use a larger sample, include additional predictors such as income, age, employment status, household size or housing costs, and compare different Austrian regions. It may also be useful to analyze more recent data to examine whether predictors of fixed-term rental agreements have changed over time.

6 References

Meyer, D., & Wurzer, M. (n.d.). Data analysis and statistical modeling. FH Technikum Wien.

Statistik Austria. (2012). Microcensus 2012: Austrian household and labor market survey.

7 Session Info

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
```

LAPACK version 3.12.1

locale:

[1] LC_COLLATE=English_Canada.utf8 LC_CTYPE=English_Canada.utf8
[3] LC_MONETARY=English_Canada.utf8 LC_NUMERIC=C
[5] LC_TIME=English_Canada.utf8

time zone: Europe/Vienna

tzcode source: internal

attached base packages:

[1] grid stats graphics grDevices utils datasets methods
[8] base

other attached packages:

[1] patchwork_1.3.2 ggplot2_4.0.2 ggeffects_2.3.2 see_0.13.0
[5] performance_0.16.0 DHARMA_0.4.7 dplyr_1.1.4 effects_4.2-4
[9] vcd_1.4-13 car_3.1-3 carData_3.0-5 pander_0.6.6

loaded via a namespace (and not attached):

[1] gtable_0.3.6 xfun_0.52 insight_1.4.4 lattice_0.22-7
[5] vctrs_0.6.5 tools_4.5.1 Rdpack_2.6.4 generics_0.1.4
[9] datawizard_1.3.0 parallel_4.5.1 tibble_3.3.0 pkgconfig_2.0.3
[13] Matrix_1.7-3 RColorBrewer_1.1-3 S7_0.2.1 lifecycle_1.0.4
[17] compiler_4.5.1 farver_2.1.2 tinytex_0.57 codetools_0.2-20
[21] gap.datasets_0.0.6 mitools_2.4 survey_4.4-8 httpuv_1.6.17
[25] htmltools_0.5.8.1 yaml_2.3.10 Formula_1.2-5 later_1.4.2
[29] pillar_1.11.0 nloptr_2.2.1 MASS_7.3-65 iterators_1.0.14
[33] reformulas_0.4.1 boot_1.3-31 abind_1.4-8 foreach_1.5.2
[37] mime_0.13 nlme_3.1-168 tidyselect_1.2.1 digest_0.6.37
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[49] withr_3.0.2 promises_1.5.0 scales_1.4.0 rmarkdown_2.29
[53] otel_0.2.0 nnet_7.3-20 lme4_1.1-37 hms_1.1.3
[57] zoo_1.8-14 shiny_1.13.0 evaluate_1.0.4 haven_2.5.5
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[65] doParallel_1.0.17 mgcv_1.9-3 rlang_1.1.6 Rcpp_1.1.0
[69] xtable_1.8-8 glue_1.8.0 qgam_2.0.0 DBI_1.2.3
[73] minqa_1.2.8 jsonlite_2.0.0 R6_2.6.1 plyr_1.8.9